



MIST- MEDICAL INSTITUTE FOR SCREENING TEST

RANDOM MOCK-1 DATED ON 5TH MAY 2026

QUESTIONS

- Q1. A 27-year-old male is brought to the emergency department after a motorcycle accident with direct trauma to his shoulder region. He complains of severe pain and difficulty in moving the affected upper limb. On examination, he is unable to abduct the arm beyond the initial 15°, although the initiation of abduction is preserved. There is localized tenderness around the proximal humerus. X-ray imaging reveals a fracture at the surgical neck of the humerus. Additionally, there may be decreased sensation over the lateral aspect of the shoulder. Which of the following nerves is most likely injured?
- Radial nerve
 - Axillary nerve
 - Musculocutaneous nerve
 - Suprascapular nerve
- Q2. A 70-year-old elderly woman presents to the emergency department after sustaining a fall at home, complaining of severe pain in the hip region and inability to stand or bear weight. On clinical examination, the affected limb appears shortened and externally rotated. Radiographic evaluation confirms a fracture of the neck of the femur. The orthopedic team explains the high risk of avascular necrosis associated with this injury due to disruption of the primary blood supply. This is particularly significant in elderly patients where vascular compromise is common. Which of the following arteries is the principal source of blood supply to the femoral head in adults?
- Obturator artery
 - Lateral circumflex femoral artery
 - Medial circumflex femoral artery
 - Inferior gluteal artery
- Q3. A 20-year-old man presents to the emergency department with repeated episodes of nosebleeds, especially from the anterior part of the nose. He reports that the bleeding is spontaneous and stops with simple pressure but recurs frequently. On anterior rhinoscopic examination, active bleeding is noted from the anteroinferior portion of the nasal septum. This region is known to contain a rich vascular network formed by anastomosis of multiple arteries. Such bleeds are commonly seen in young individuals due to trauma or mucosal irritation. Which of the following vascular plexuses is most likely responsible for this condition?
- Woodruff's plexus
 - Kiesselbach's plexus
 - Pterygoid plexus
 - Cavernous plexus
- Q4. A patient presents with complaints of persistent nasal obstruction, mouth breathing, and recurrent episodes of sinus infections. Symptoms are more pronounced during sleep, with associated snoring and difficulty in nasal airflow. On further evaluation, CT imaging in the sagittal plane reveals enlargement of lymphoid tissue situated in the roof and posterior wall of the nasopharynx. This hypertrophied tissue is part of the lymphatic ring guarding the upper respiratory tract. Such enlargement commonly leads to blockage of the nasopharyngeal airway and Eustachian tube dysfunction. Which of the following structures is most likely involved in this condition?
- Palatine tonsil
 - Pharyngeal tonsil (adenoids)
 - Lingual tonsil
 - Tubal tonsil



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- Q5. A 28-year-old patient presents with complaints of facial pain, nasal congestion, and purulent nasal discharge suggestive of sinusitis. The pain is localized over the cheek region and worsens on bending forward. CT scan of the paranasal sinuses reveals involvement of the largest sinus situated within the maxillary bone. This sinus is commonly affected due to its anatomical drainage pattern. The bone containing this air-filled cavity is classified as a pneumatic bone. Which of the following bones is considered pneumatic because it contains this sinus?
- Zygomatic bone
 - Maxilla
 - Palatine bone
 - Nasal bone
- Q6. A 22-year-old athlete presents to the emergency department with wrist pain following a fall on an outstretched hand. He complains of tenderness on the lateral aspect of the wrist, especially during movement. On examination, there is marked tenderness in the anatomical snuffbox region. X-ray imaging suggests a fracture involving one of the carpal bones commonly injured in such trauma. This bone has a precarious blood supply, making it prone to complications like avascular necrosis. Which of the following carpal bones is most likely fractured?
- Lunate
 - Scaphoid
 - Capitate
 - Triquetrum
- Q7. A patient presents following trauma to the shoulder with a fracture involving the surgical neck of the humerus. He complains of difficulty in performing lateral (external) rotation of the arm. On examination, there is weakness of shoulder movements along with possible sensory loss over the lateral shoulder region. The injury is suggestive of axillary nerve involvement, which supplies specific muscles of the shoulder. Among the muscles arising from the scapula, one plays a key role in lateral rotation and is affected in this condition. Which of the following muscles is most likely involved?
- Teres major
 - Teres minor
 - Subscapularis
 - Pectoralis major
- Q8. A histological specimen obtained from a joint surface is examined under the microscope. It shows chondrocytes lying within lacunae, surrounded by a smooth, homogeneous, glassy extracellular matrix. Notably, there is an absence of perichondrium covering this cartilage. The tissue is adapted to withstand compressive forces and provide a smooth, low-friction surface for joint movement. These features are characteristic of specialized cartilage found at synovial joint surfaces. Which of the following types of cartilage best corresponds to these findings?
- Hyaline cartilage of trachea
 - Elastic cartilage of ear
 - Articular cartilage
 - Fibrocartilage
- Q9. A patient presenting with features of renal tubular dysfunction undergoes a kidney biopsy for further evaluation. On histological examination, the renal tubules are seen lined by cells that are cuboidal in shape with centrally placed round nuclei. These cells are arranged in a single layer and are specialized for absorption and secretion. Such epithelial lining is characteristic of most segments of the renal tubules, especially the proximal and distal convoluted tubules. Which of the following types of epithelium lines the renal tubules?



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- a) Simple squamous epithelium
- b) Simple cuboidal epithelium
- c) Stratified cuboidal epithelium
- d) Transitional epithelium

Q10. A 48-year-old chronic smoker presents with complaints of persistent cough and recurrent episodes of respiratory infections. He has a long history of exposure to cigarette smoke and airway irritation. Bronchial biopsy reveals that the normal pseudostratified ciliated columnar epithelium has been replaced by stratified squamous epithelium. This adaptive change occurs in response to chronic irritation and helps the tissue withstand stress, but compromises mucociliary clearance. Over time, this process may predispose to dysplasia and malignancy. Which histological process is responsible for this epithelial change?

- a) Dysplasia
- b) Metaplasia
- c) Hyperplasia
- d) Hypertrophy

Q11. A newborn presents immediately after birth with severe respiratory distress and cyanosis. On examination, breath sounds are diminished on the left side, and the abdomen appears scaphoid. Chest X-ray demonstrates herniation of intestinal loops into the left thoracic cavity with mediastinal shift. The clinical picture is suggestive of a Bochdalek congenital diaphragmatic hernia, most commonly occurring on the left side. This condition results from a defect in the formation of a key embryological component of the diaphragm. Failure of development of which of the following structures is most likely responsible for this condition?

- a) Septum transversum
- b) Pleuroperitoneal membrane
- c) Dorsal mesentery of esophagus
- d) Somatic mesoderm of body wall

Q12. A newborn presents with bilious vomiting soon after birth along with signs of intestinal obstruction. On examination, there is abdominal distension and failure to pass meconium. Imaging studies reveal abnormal positioning of intestinal loops with twisting around the superior mesenteric artery. This suggests a defect in the normal developmental process of the midgut. Normally, the midgut undergoes physiological herniation followed by a 270° counterclockwise rotation before returning to the abdominal cavity. Which of the following embryological processes is most likely defective in this condition?

- a) Gastrulation
- b) Physiological herniation and rotation of midgut
- c) Neurulation
- d) Septum transversum development

Q13. A researcher examining early human embryonic development focuses on changes occurring during the second week after fertilization. At this stage, the embryonic structure organizes into a distinct bilaminar germ disc. This disc consists of two cellular layers that play a crucial role in further differentiation and formation of the trilaminar embryo in the subsequent weeks. These layers are derived from the inner cell mass and are essential for early developmental processes. Which of the following layers constitute the bilaminar germ disc?

- a) Ectoderm and mesoderm
- b) Epiblast and hypoblast
- c) Mesoderm and endoderm
- d) Endoderm and ectoderm



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- Q14. A 40-year-old man presents with sudden onset severe pain in the posterior aspect of the ankle while playing tennis. He describes a “snap” sensation followed by difficulty in walking. On examination, he is unable to stand on his toes, indicating loss of plantar flexion. Ultrasound confirms rupture of the calcaneal (Achilles) tendon. This tendon is the strongest tendon in the body and is formed by the fusion of muscles responsible for powerful plantar flexion. Which of the following muscle groups primarily contributes to the formation of this tendon?
- Tibialis posterior and flexor digitorum longus
 - Gastrocnemius and soleus
 - Fibularis longus and brevis
 - Tibialis anterior and extensor digitorum longus
- Q15. A 22-year-old football player presents with acute knee pain and a feeling of instability after a twisting injury during play. He reports difficulty in bearing weight and a sensation of the knee “giving way.” On clinical examination, the anterior drawer test is positive, indicating excessive forward movement of the tibia relative to the femur. This suggests damage to a key stabilizing ligament of the knee joint that prevents anterior displacement of the tibia. Such injuries are common in sports involving sudden deceleration and pivoting movements. Which of the following ligaments is most likely injured?
- Posterior cruciate ligament
 - Anterior cruciate ligament
 - Medial collateral ligament
 - Lateral collateral ligament
- Q16. A 58-year-old man with chronic kidney disease presents with progressive weakness. His serum potassium is markedly elevated. A physiologist explains that the excitability of skeletal muscle initially changes because hyperkalemia alters the resting membrane potential of excitable cells. Which of the following best describes the immediate effect of moderate hyperkalemia on the resting membrane potential?
- It becomes more negative due to increased Na^+ influx
 - It becomes less negative due to reduced K^+ gradient across the membrane
 - It becomes unchanged because Na^+/K^+ ATPase fully compensates
 - It becomes more negative due to increased Cl^- conductance
- Q17. A 30-year-old research volunteer participates in a pharmacological study and is administered a drug that blocks voltage-gated sodium channels in peripheral nerves. Shortly after administration, he develops loss of nerve impulse conduction with failure of action potential propagation. These sodium channels are essential for the rapid influx of sodium ions during nerve excitation. Blocking them disrupts the phase responsible for the rising portion of the action potential. Which phase of the action potential is primarily affected by this drug?
- Resting phase
 - Depolarization phase
 - Repolarization phase
 - After-hyperpolarization phase
- Q18. A patient diagnosed with Fanconi syndrome presents with features of generalized proximal tubular dysfunction. He complains of polyuria, dehydration, and weakness due to loss of essential solutes in urine. Laboratory evaluation reveals multiple substances being abnormally excreted despite normal plasma levels. The proximal tubule is responsible for reabsorption of glucose, amino acids, bicarbonate, and phosphate. Defective reabsorption leads to inappropriate urinary loss of these substances. Which of the following substances is most likely to appear in the urine in this condition?
- Glucose



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- b) Urea
- c) Creatinine
- d) Inulin

Q19. A 55-year-old patient undergoes renal function testing using para-aminohippuric acid (PAH) infusion to assess kidney perfusion. PAH is freely filtered at the glomerulus and is also actively secreted by the proximal tubules, leading to nearly complete clearance from plasma in a single passage through the kidneys. Because of this property, its clearance reflects the volume of plasma flowing through the renal vasculature. This makes it a useful tool for evaluating renal hemodynamics. Which of the following renal parameters is measured by PAH clearance?

- a) Glomerular filtration rate
- b) Filtration fraction
- c) Renal plasma flow
- d) Renal blood volume

Q20. A 65-year-old man presents with episodes of dizziness and syncope. ECG reveals absence of normal atrial depolarization and a slow ventricular rhythm (~40 beats/min). Further evaluation suggests failure of the primary cardiac pacemaker. Which structure normally acts as the primary pacemaker of the heart?

- a) Atrioventricular node
- b) Sinoatrial node
- c) Purkinje fibers
- d) Bundle of His

Q21. A 62-year-old man with a long history of smoking presents with progressive shortness of breath and decreased lung compliance. He reports worsening dyspnea on exertion over several months. Histological examination of lung tissue reveals damage to the cells responsible for producing surfactant, leading to increased alveolar surface tension and impaired lung expansion. These cells play a crucial role in maintaining alveolar stability and preventing collapse during respiration. Which of the following lung cell types is primarily responsible for surfactant production?

- a) Type I pneumocytes
- b) Type II pneumocytes
- c) Alveolar macrophages
- d) Clara cells

Q22. A 25-year-old medical student studying neurophysiology focuses on how the body processes sensory information and produces coordinated motor responses. He learns that this function is carried out by a highly organized system responsible for integration, interpretation, and response generation. This system serves as the main control center of the body, receiving inputs from peripheral receptors and sending outputs to effectors. It consists of two major anatomical components enclosed within the skull and vertebral column. Which of the following structures together constitute the central nervous system?

- a) Brain and spinal nerves
- b) Brain and spinal cord
- c) Brainstem and cranial nerves
- d) Spinal cord and peripheral nerves

Q23. A patient with a spinal cord injury presents with loss of voluntary motor control below the level of the lesion. However, on examination, certain involuntary movements such as withdrawal reflexes are still preserved. This indicates that some neural circuits remain functional despite loss of higher control from the brain. The spinal cord is capable of independently processing specific neural pathways responsible for these



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automatic responses. Such functions highlight its role in basic neural integration. The spinal cord in this scenario is functioning primarily as a:

- a) Sensory receptor
- b) Reflex center
- c) Endocrine organ
- d) Neurotransmitter storage site

Q24. A patient with long-standing Chronic Obstructive Pulmonary Disease presents with chronic hypercapnia and adaptation of central chemoreceptors. Due to persistently elevated CO_2 levels, the central drive to respiration becomes blunted over time. As a result, ventilation is no longer primarily stimulated by increased PaCO_2 . Instead, peripheral chemoreceptors in the carotid and aortic bodies become the main regulators of breathing. These receptors respond predominantly to changes in arterial oxygen levels. Which of the following stimuli primarily drives respiration in this patient?

- a) Increased PaCO_2
- b) Decreased PaO_2
- c) Increased blood Ph
- d) Increased bicarbonate

Q25. A 50-year-old man undergoes abdominal surgery and later develops abdominal distension, vomiting, and absence of bowel movements. Clinical findings suggest postoperative intestinal ileus with loss of peristalsis. The condition is attributed to disruption of the intrinsic neural network that regulates coordinated smooth muscle activity in the gastrointestinal tract. This neural plexus lies between the longitudinal and circular muscle layers and is primarily responsible for controlling gut motility. Which of the following structures plays the major role in regulating gastrointestinal smooth muscle activity?

- a) Vagus nerve
- b) Myenteric (Auerbach) plexus
- c) Submucosal (Meissner) plexus
- d) Sympathetic ganglia

Q26. A patient who has undergone vagotomy presents with reduced gastric motility and delayed emptying of the stomach. The vagus nerve normally provides parasympathetic innervation to the gastrointestinal tract, promoting smooth muscle contraction. Loss of this input leads to decreased peristalsis and impaired digestive function. Parasympathetic nerve endings release a key neurotransmitter that directly stimulates contraction of GI smooth muscle. Which of the following neurotransmitters is primarily responsible for this effect?

- a) Norepinephrine
- b) Acetylcholine
- c) Dopamine
- d) Serotonin

Q27. A 58-year-old man with long-standing diabetes presents with excessive urination and increased thirst. Laboratory evaluation reveals high plasma osmolality with inappropriately dilute urine, suggesting impaired water reabsorption. Following administration of desmopressin, there is a marked increase in urine osmolality, indicating responsiveness of renal tubules to antidiuretic hormone. Desmopressin acts on V_2 receptors in the collecting ducts, promoting water reabsorption. This effect is mediated by insertion of specific water channels into the tubular membrane. Which of the following physiological mechanisms best explains this response?

- a) Increased sodium reabsorption in proximal tubule
- b) Increased aquaporin-2 insertion in collecting ducts
- c) Increased glomerular filtration rate
- d) Increased urea secretion in distal tubule



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- Q28. A patient with a pituitary tumor presents with features of hyperprolactinemia, including galactorrhea, menstrual irregularities, and infertility. Laboratory evaluation confirms elevated serum prolactin levels. Under normal physiological conditions, prolactin secretion from the anterior pituitary is under inhibitory control by a hypothalamic factor. Loss of this inhibitory influence or tumor-related dysregulation leads to excessive prolactin secretion. This regulatory mechanism is unique among pituitary hormones as it is predominantly inhibitory. Which of the following hypothalamic hormones normally suppresses prolactin secretion?
- TRH
 - Dopamine
 - GnRH
 - CRH
- Q29. A patient with Type 1 Diabetes Mellitus presents with polyuria, polydipsia, and weight loss due to absolute insulin deficiency. Laboratory findings reveal hyperglycemia with low circulating insulin levels. This condition results from autoimmune destruction of specific cells in the pancreatic islets of Langerhans. These cells are responsible for synthesis and secretion of insulin, which is essential for glucose uptake and metabolism. Loss of their function leads to impaired glucose utilization and metabolic imbalance. Which of the following pancreatic cells normally secretes insulin?
- Alpha cells
 - Beta cells
 - Delta cells
 - PP cells
- Q30. A patient presents with progressive difficulty in seeing in dim light and complains of poor vision at night. He reports frequent stumbling in low-light conditions, suggesting impairment of scotopic vision. Ophthalmic examination reveals degeneration of photoreceptor cells specialized for low-light sensitivity. These cells are highly sensitive to light and are predominantly located in the peripheral retina. Their dysfunction leads to night blindness and reduced dark adaptation. Which of the following photoreceptors is primarily affected in this condition?
- Cones
 - Rods
 - Bipolar cells
 - Horizontal cells
- Q31. A 58-year-old man presents to the emergency department with severe retrosternal chest pain radiating to the left arm, persisting for the past 2 hours. He also reports associated sweating and shortness of breath. ECG demonstrates ST-segment elevation in the anterior leads, suggestive of acute myocardial infarction. To confirm myocardial injury, cardiac biomarkers are evaluated. Among these, one marker is highly specific and sensitive, especially in the early phase of myocardial damage. Which of the following enzymes/markers is most specific for early diagnosis of myocardial infarction?
- Creatine kinase-MB
 - Lactate dehydrogenase-1
 - Troponin-I
 - Aspartate aminotransferase
- Q32. A 7-year-old boy is brought with a history of developmental delay, self-mutilating behavior such as lip and finger biting, and episodes of joint pain suggestive of gout. Parents also report the presence of orange-colored crystals in the diaper. Laboratory evaluation reveals markedly elevated serum uric acid levels. The clinical picture is characteristic of a purine metabolism disorder caused by deficiency of a key salvage



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pathway enzyme. This leads to overproduction of uric acid and neurological manifestations. Which of the following enzyme deficiencies is most likely responsible for this condition?

- a) Xanthine oxidase
- b) Hypoxanthine-guanine phosphoribosyltransferase (HGPRT)
- c) Adenosine deaminase
- d) Carbamoyl phosphate synthetase II

Q33. A researcher is analyzing the kinetic properties of an enzyme in the presence of a specific drug. It is observed that upon drug binding, the enzyme shows a decrease in K_m , indicating increased apparent affinity for the substrate, without altering the maximum reaction velocity (V_{max}). This type of inhibition occurs when the inhibitor binds only to the enzyme-substrate complex. As a result, both enzyme activity and substrate binding are affected in a characteristic manner. Which of the following types of enzyme inhibition best explains this observation?

- a) Competitive inhibition
- b) Non-competitive inhibition
- c) Uncompetitive inhibition
- d) Irreversible inhibition

Q34. A 32-year-old man is brought to the emergency department after exposure to smoke in a house fire. He presents with headache, dizziness, confusion, and signs of hypoxia. Laboratory investigations reveal elevated levels of carboxyhemoglobin, indicating carbon monoxide poisoning. Carbon monoxide has a high affinity for hemoglobin and also interferes with cellular respiration by inhibiting a key component of the electron transport chain. This leads to impaired oxidative phosphorylation and decreased ATP production. Which component of the electron transport chain is primarily inhibited in this condition?

- a) Complex I – NADH dehydrogenase
- b) Complex II – Succinate dehydrogenase
- c) Complex III – Cytochrome bc1 complex
- d) Complex IV – Cytochrome oxidase

Q35. A researcher is evaluating the effect of a novel drug on mitochondrial function. It is observed that the drug disrupts the proton gradient across the inner mitochondrial membrane without inhibiting the electron transport chain. As a result, electrons continue to flow through the complexes, but ATP synthesis is impaired. This leads to dissipation of energy as heat rather than being stored as ATP. Such a mechanism indicates separation of electron transport from ATP production. Which of the following mechanisms best explains the action of this drug?

- a) Competitive inhibition of ATP synthase
- b) Uncoupling of oxidative phosphorylation
- c) Blocking electron transfer at Complex III
- d) Inhibition of NADH oxidation

Q36. A 2-year-old child is brought to the clinic with progressive abdominal distension and failure to thrive. On examination, hepatomegaly is noted, and laboratory findings reveal fasting hypoglycemia, lactic acidosis, and hypertriglyceridemia. Further evaluation shows excessive accumulation of glycogen in the liver. These features are characteristic of a glycogen storage disorder affecting glucose release from the liver. The defect lies in the final step of gluconeogenesis and glycogenolysis, leading to impaired conversion of glucose-6-phosphate to free glucose. Which of the following enzyme deficiencies is most likely responsible for this condition?

- a) Glycogen phosphorylase
- b) Glucose-6-phosphatase



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- c) Branching enzyme
- d) Debranching enzyme

Q37. A researcher studying red blood cell metabolism observes that erythrocytes lack mitochondria and therefore rely entirely on glucose for their energy needs. In addition to ATP production, RBCs require a continuous supply of NADPH to protect against oxidative damage. NADPH is essential for maintaining glutathione in its reduced form, which prevents hemolysis. This reducing power is generated through a specific metabolic pathway active in RBCs. Which of the following pathways is the primary source of NADPH in red blood cells?

- a) Glycolysis
- b) TCA cycle
- c) Pentose phosphate pathway
- d) Glycogenolysis

Q38. A patient undergoing prolonged fasting maintains adequate blood glucose levels despite lack of dietary intake. During this state, glycogen stores become depleted, and the liver shifts to producing glucose from non-carbohydrate precursors such as lactate, glycerol, and amino acids. This process is essential for supplying glucose to glucose-dependent tissues like the brain and red blood cells. It represents a key adaptive metabolic pathway activated during starvation. Which of the following pathways is primarily responsible for maintaining blood glucose in this condition?

- a) Glycolysis
- b) Glycogenesis
- c) Gluconeogenesis
- d) Pentose phosphate pathway

Q39. A 45-year-old man with Type 2 Diabetes Mellitus presents with persistent hyperglycemia despite lifestyle modifications. He is started on a medication that activates AMP-activated protein kinase (AMPK), leading to decreased hepatic glucose output. This drug reduces the liver's ability to generate glucose from non-carbohydrate sources. As a result, fasting blood glucose levels improve. The primary metabolic effect is suppression of a key glucose-producing pathway in the liver. Which of the following metabolic pathways is mainly inhibited by this drug?

- a) Glycolysis
- b) Glycogenesis
- c) Gluconeogenesis
- d) Pentose phosphate pathway

Q40. A 19-year-old athlete collapses during intense physical activity and is found to have severe hypoglycemia with elevated lactate levels. Further evaluation suggests a defect in glycolysis, particularly involving a key regulatory enzyme, phosphofructokinase-1 (PFK-1). This enzyme plays a crucial role in controlling the rate of glycolysis and is considered the rate-limiting step. It catalyzes an important phosphorylation reaction early in the pathway, committing glucose to further metabolism. Which of the following steps in glycolysis is catalyzed by PFK-1?

- a) Glucose → Glucose-6-phosphate
- b) Fructose-6-phosphate → Fructose-1,6-bisphosphate
- c) Glyceraldehyde-3-phosphate → 1,3-bisphosphoglycerate
- d) Pyruvate → Lactate

Q41. A 3-year-old child is brought with developmental delay, recurrent seizures, and hypopigmented skin. The parents also report a characteristic musty or "mousy" odor of the urine. Laboratory investigations reveal



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markedly elevated phenylalanine levels, suggesting a defect in phenylalanine metabolism consistent with phenylketonuria. In this condition, the conversion of phenylalanine to another amino acid is impaired. As a result, that amino acid must be obtained from the diet and becomes conditionally essential. Which of the following amino acids becomes conditionally essential in this disorder?

- a) Tyrosine
- b) Tryptophan
- c) Methionine
- d) Histidine

Q42. A newborn presents within 48 hours of birth with vomiting, lethargy, poor feeding, and signs of respiratory alkalosis. Laboratory evaluation reveals severe hyperammonemia, raising suspicion of a urea cycle disorder. Further testing confirms deficiency of ornithine transcarbamylase, an enzyme responsible for conversion of carbamoyl phosphate and ornithine into citrulline. Due to this enzymatic block, substrates proximal to the defect accumulate in excess. This leads to toxic buildup of ammonia and related intermediates. Which of the following intermediates is most likely to accumulate in this condition?

- a) Citrulline
- b) Carbamoyl phosphate
- c) Argininosuccinate
- d) Ornithine

Q43. A 3-year-old child is brought with developmental delay, recurrent seizures, and hypopigmented skin. Parents also report a characteristic musty or “mousy” odor of the urine. Laboratory investigations reveal markedly elevated levels of phenylalanine in the blood. The clinical features are suggestive of phenylketonuria, a disorder of amino acid metabolism. This condition results from deficiency of a key enzyme responsible for converting phenylalanine to tyrosine. Which of the following enzyme deficiencies is most likely responsible for this condition?

- a) Tyrosinase
- b) Phenylalanine hydroxylase
- c) Homogentisate oxidase
- d) Tyrosine aminotransferase

Q44. A teenage patient presents with features of tall stature, long limbs, and lens dislocation, along with a history of recurrent thromboembolic events. Clinical examination also reveals skeletal abnormalities resembling Marfan-like features. Laboratory investigations demonstrate markedly elevated homocysteine levels in the blood. This suggests a defect in the transsulfuration pathway of methionine metabolism. The condition is classically caused by deficiency of an enzyme required for conversion of homocysteine to cystathionine. Which of the following enzyme defects is most likely responsible for this condition?

- a) Methionine synthase
- b) Cystathionine β -synthase
- c) Homogentisate oxidase
- d) Tyrosine aminotransferase

Q45. A patient with uncontrolled Diabetes Mellitus presents with features of ketoacidosis, including dehydration, abdominal pain, and altered sensorium. Laboratory findings reveal elevated levels of ketone bodies in blood and urine. In this condition, increased fatty acid oxidation leads to excess production of acetyl-CoA. The liver converts this excess acetyl-CoA into ketone bodies, which serve as an alternative energy source during insulin deficiency. This metabolic process occurs primarily in a specific organ. Which of the following organs is responsible for the synthesis of ketone bodies?

- a) Brain



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- b) Muscle
- c) Liver
- d) Kidney

Q46. A 58-year-old man with long-standing hypertension presents with acute onset severe chest pain followed by collapse. ECG reveals ST-segment elevation in the anterior leads, and angiography confirms complete occlusion of the left anterior descending artery. Within 30 minutes of ischemia, myocardial cells exhibit cellular swelling and reversible mitochondrial dysfunction. At this early stage, the injury is still potentially reversible if blood flow is restored. The primary underlying event is failure of energy production due to impaired oxidative metabolism. Which of the following mechanisms represents the earliest biochemical event responsible for reversible cell injury in this condition?

- a) Increased lysosomal membrane permeability
- b) ATP depletion due to oxidative phosphorylation failure
- c) Activation of caspases
- d) Nuclear chromatin fragmentation

Q47. A patient presents with features of anemia along with fatigue, pallor, and increased susceptibility to infections. Laboratory evaluation reveals normocytic normochromic red blood cells with a markedly low reticulocyte count, indicating decreased bone marrow activity. In addition, there is pancytopenia involving reduction in red cells, white cells, and platelets. These findings suggest a primary failure of bone marrow hematopoiesis rather than peripheral destruction or deficiency. Which of the following conditions best explains this clinical presentation?

- a) Iron deficiency anemia
- b) Hemolytic anemia
- c) Aplastic anemia
- d) Megaloblastic anemia

Q48. A 36-year-old man presents with chronic cough, weight loss, and persistent low-grade fever. He also reports night sweats and general malaise over the past few months. Chest X-ray reveals cavitory lesions in the upper lobe of the lung. Histopathological examination of lung biopsy shows granulomas with central caseous necrosis surrounded by epithelioid cells and Langhans giant cells. These findings are characteristic of a chronic granulomatous infection. Which of the following organisms is most commonly responsible for this pathology?

- a) *Mycobacterium tuberculosis*
- b) *Staphylococcus aureus*
- c) *Streptococcus pneumoniae*
- d) *Pseudomonas aeruginosa*

Q49. A 14-year-old boy presents with jaundice, splenomegaly, and mild anemia. Peripheral blood smear reveals spherocytes, and the osmotic fragility test is positive. There is a significant family history of similar symptoms, suggesting a hereditary disorder. The findings indicate increased destruction of red blood cells due to structural abnormalities. This condition is typically caused by defects in proteins that maintain the integrity and shape of the RBC membrane. Which of the following defects is most likely responsible for this condition?

- a) Defect in hemoglobin synthesis
- b) Defect in RBC membrane cytoskeleton proteins
- c) Defect in folate metabolism
- d) Defect in iron absorption



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- Q50. A 58-year-old man with acute myocardial infarction undergoes urgent reperfusion therapy. Soon after restoration of blood flow, there is worsening of myocardial injury due to generation of reactive oxygen species (ROS). These free radicals cause cellular damage by attacking essential components of the cell. One of the most significant effects is injury to cell membranes, leading to increased permeability and cell dysfunction. This process plays a major role in reperfusion injury. Which of the following mechanisms most directly explains tissue damage in this condition?
- a) DNA methylation
 - b) Lipid peroxidation of cell membranes
 - c) Increased glycogen synthesis
 - d) Protein glycosylation
- Q51. A 48-year-old man presents with progressive fatigue, unintentional weight loss, and marked splenomegaly. Laboratory investigations reveal significant leukocytosis with presence of myeloid cells at various stages of maturation. Peripheral smear and bone marrow findings suggest a chronic myeloproliferative disorder. Cytogenetic analysis demonstrates the presence of a t(9;22) translocation, also known as the Philadelphia chromosome. This genetic abnormality results in formation of an abnormal fusion gene with constitutive tyrosine kinase activity. What is the underlying molecular abnormality in this condition?
- a) BCL-2 overexpression
 - b) PML-RARA fusion
 - c) BCR-ABL fusion
 - d) c-MYC activation
- Q52. A patient presents with mucocutaneous bleeding, petechiae, and easy bruising, suggestive of a platelet disorder. Laboratory evaluation reveals isolated thrombocytopenia with normal coagulation parameters. The clinical picture is consistent with immune thrombocytopenic purpura (ITP), an autoimmune condition characterized by destruction of platelets. In this disorder, autoantibodies are directed against specific glycoproteins present on the platelet surface, leading to their premature removal by the spleen. Which of the following platelet components is most commonly targeted by autoantibodies in this condition?
- a) GpIb
 - b) GpIIb/IIIa
 - c) Factor VIII
 - d) Fibrinogen
- Q53. A 35-year-old woman develops excessive bleeding following delivery complicated by placental abruption. She presents with oozing from multiple sites and signs of hemodynamic instability. Laboratory investigations reveal prolonged PT and aPTT, decreased platelet count, and markedly elevated D-dimer levels. These findings indicate widespread activation of the coagulation cascade with simultaneous consumption of clotting factors and platelets. This results in both thrombosis and bleeding manifestations. Which of the following is the most likely diagnosis in this patient?
- a) Hemophilia A
 - b) von Willebrand disease
 - c) Disseminated intravascular coagulation
 - d) Immune thrombocytopenic purpura
- Q54. A 45-year-old patient presents with signs of acute bacterial infection, including fever and localized inflammation. Laboratory findings show elevated neutrophil count, and histological examination demonstrates accumulation of neutrophils at the site of infection. During the inflammatory response, leukocytes first adhere to the vascular endothelium and then migrate out of the bloodstream into surrounding tissues. This step



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involves passage of leukocytes through intercellular junctions of endothelial cells. Which of the following processes describes this movement of leukocytes into tissues?

- a) Margination
- b) Diapedesis
- c) Opsonization
- d) Phagocytosis

Q55. A 5-year-old child is brought with a rapidly growing red lesion on the face, noted since early childhood. The lesion appears bright red, slightly raised, and increases in size over a short period. There is no associated pain or ulceration. Over time, the parents observe gradual spontaneous regression of the lesion without any treatment. This pattern of rapid growth followed by involution is characteristic of a benign vascular tumor seen in children. Which of the following is the most likely diagnosis?

- a) Angiosarcoma
- b) Hemangioma
- c) Kaposi sarcoma
- d) Glomus tumor

Q56. A 38-year-old woman presents with fever, malaise, and productive cough suggestive of bacterial pneumonia. Laboratory investigations reveal leukocytosis along with markedly elevated serum C-reactive protein (CRP) levels, indicating an acute-phase response. During inflammation, certain cytokines stimulate the liver to produce acute-phase proteins such as CRP. Among these, one cytokine plays a central role in mediating hepatic synthesis of CRP. Which of the following cytokines is primarily responsible for stimulating CRP production in this condition?

- a) Interleukin-2
- b) Interleukin-6
- c) Interleukin-10
- d) Interferon- γ

Q57. A patient presents with symptoms of a parasitic infection and laboratory findings reveal marked eosinophilia. On further evaluation, the immune response is found to be directed against helminths and other parasites. Certain immune cells are specifically adapted to combat parasitic infections by releasing cytotoxic granules and mediating antibody-dependent cellular cytotoxicity. These cells play a crucial role in host defense against large extracellular parasites. Which of the following immune cells is primarily responsible for defense against parasitic infections?

- a) Eosinophils
- b) Basophils
- c) Plasma cells
- d) Dendritic cells

Q58. A 35-year-old man with viral hepatitis develops early immune-mediated destruction of infected hepatocytes. This response occurs rapidly, even before the activation of antigen-specific adaptive immunity. The cells responsible are part of the innate immune system and can recognize and kill virus-infected cells without prior sensitization. They act through detection of reduced MHC class I expression and release cytotoxic granules. Which of the following immune cells is primarily responsible for this early antiviral defense?

- a) Cytotoxic T lymphocytes
- b) B lymphocytes
- c) Natural killer cells
- d) Plasma cells



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- Q59. A child presents with developmental delay, hypotonia, and feeding difficulties in early infancy. As the child grows, features such as hyperphagia and obesity may develop. Genetic analysis reveals a deletion on chromosome 15 that is inherited from the father. This pattern of inheritance is associated with genomic imprinting, where the paternal gene expression is lost. The clinical picture is characteristic of a specific genetic syndrome involving hypothalamic dysfunction. Which of the following syndromes is most likely responsible for this condition?
- Angelman syndrome
 - Prader–Willi syndrome
 - Turner syndrome
 - Edwards syndrome
- Q60. A pedigree analysis reveals that all offspring of affected mothers inherit the disease, whereas affected fathers do not transmit the condition to any of their children. The pattern shows maternal transmission across generations without paternal contribution. This suggests inheritance through cytoplasmic organelles rather than nuclear genes. Such disorders are typically associated with mutations in mitochondrial DNA, which is exclusively inherited from the mother. Which of the following inheritance patterns best explains this observation?
- Autosomal dominant
 - X-linked recessive
 - Mitochondrial inheritance
 - Autosomal recessive
- Q61. A 28-year-old farmer is brought to the emergency department after exposure to organophosphate pesticides. He presents with excessive salivation, sweating, miosis (pinpoint pupils), muscle fasciculations, bradycardia, and hypotension. These features indicate cholinergic toxicity due to inhibition of acetylcholinesterase, leading to accumulation of acetylcholine at synapses. Both muscarinic and nicotinic receptors are overstimulated, causing widespread symptoms. Management requires a drug that can reactivate acetylcholinesterase and reverse effects at both receptor types. Which of the following drugs should be administered in this condition?
- Atropine
 - Pralidoxime
 - Glycopyrrolate
 - Physostigmine
- Q62. A patient undergoing chemotherapy develops symptoms of dysuria, hematuria, and bladder irritation. Further evaluation suggests hemorrhagic cystitis as a complication of treatment. This condition is caused by a toxic metabolite called acrolein, which accumulates in the urinary bladder and damages the urothelium. The offending drug is an alkylating agent commonly used in various malignancies. Prophylactic administration of mesna is often used to prevent this toxicity. Which of the following drugs is most likely responsible for this adverse effect?
- Methotrexate
 - Cyclophosphamide
 - Vincristine
 - Paclitaxel
- Q63. A patient receiving chemotherapy with methotrexate develops severe mucositis due to toxicity in rapidly dividing normal cells. Methotrexate inhibits dihydrofolate reductase (DHFR), leading to reduced synthesis of tetrahydrofolate and impaired DNA synthesis. To counteract this effect, folinic acid (leucovorin)



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is administered as rescue therapy. Folinic acid provides a reduced form of folate that can directly participate in nucleotide synthesis without requiring DHFR. This helps protect normal cells while allowing the antitumor effect to continue. What is the mechanism of action of folinic acid in this context?

- a) Activates DHFR
- b) Bypasses DHFR inhibition
- c) Inhibits thymidylate synthase
- d) Activates DNA polymerase

Q64. A 52-year-old man with a history of hypertension and Type 2 Diabetes Mellitus presents for routine follow-up. He is found to have elevated blood pressure of 150/95 mmHg along with microalbuminuria on laboratory testing. The presence of microalbuminuria indicates early diabetic nephropathy. In such patients, the preferred antihypertensive agents are those that provide renal protection by reducing intraglomerular pressure and slowing disease progression. Which of the following drugs is most appropriate for this patient?

- a) Amlodipine
- b) Atenolol
- c) Enalapril
- d) Hydralazine

Q65. A 50-year-old man with a history of prolonged NSAID use presents with epigastric pain and dyspeptic symptoms. The clinical picture suggests NSAID-induced gastric mucosal injury due to reduced prostaglandin synthesis. He is started on a medication that acts directly on gastric parietal cells to suppress acid secretion. This drug irreversibly inhibits the H^+/K^+ ATPase (proton pump), leading to profound and long-lasting reduction in gastric acid production. It is commonly used in the management of peptic ulcer disease and gastroesophageal reflux. Which of the following drugs is most likely prescribed in this case?

- a) Ranitidine
- b) Omeprazole
- c) Sucralfate
- d) Misoprostol

Q66. A 7-year-old child presents with frequent episodes of brief staring spells lasting 10–15 seconds, with no postictal confusion. EEG shows generalized 3 Hz spike-and-wave discharges. He is started on a drug that blocks T-type calcium channels. Which drug is most appropriate?

- a) Valproate
- b) Ethosuximide
- c) Carbamazepine
- d) Phenytoin

Q67. A patient with persistent chronic cough is prescribed codeine for symptomatic relief. Codeine is an opioid that acts centrally within the central nervous system. It exerts its antitussive effect by directly depressing the cough reflex pathways. Specifically, it reduces the sensitivity of the cough center located in the medulla, thereby decreasing the frequency and intensity of coughing. This mechanism makes it effective in controlling non-productive cough. Which of the following mechanisms best explains the antitussive action of codeine?

- a) Peripheral bronchial dilation
- b) Suppression of medullary cough center
- c) Increased mucus clearance
- d) Histamine blockade



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- Q68. A patient with unstable angina is started on antiplatelet therapy to prevent further thrombus formation. The drug administered acts by blocking glycoprotein IIb/IIIa receptors on platelets, thereby inhibiting fibrinogen binding and preventing platelet aggregation. This mechanism directly interferes with the final common pathway of platelet aggregation. Such agents are commonly used in acute coronary syndromes and during percutaneous coronary interventions. Which of the following drugs works by this mechanism?
- Aspirin
 - Ticagrelor
 - Abciximab
 - Dipyridamole
- Q69. A patient receiving insulin therapy presents with sweating, confusion, palpitations, and tachycardia, suggestive of hypoglycemia. Normally, the body counteracts hypoglycemia through release of counter-regulatory hormones. Among these, one hormone plays the most immediate and critical role by promoting hepatic glucose production via glycogenolysis and gluconeogenesis. Deficiency of this hormone impairs recovery from hypoglycemia and prolongs its effects. Which of the following hormone deficiencies is primarily responsible for impaired recovery in this condition?
- Glucagon
 - Cortisol
 - Growth hormone
 - Epinephrine
- Q70. A 30-year-old woman presents with recurrent episodes of migraine characterized by unilateral throbbing headache, nausea, and photophobia. She is prescribed sumatriptan, a drug commonly used for acute migraine attacks. This medication acts by stimulating specific serotonin receptors located on cranial blood vessels, leading to vasoconstriction and inhibition of neuropeptide release. This mechanism helps relieve migraine symptoms by reducing vascular dilation and inflammation. Which of the following serotonin receptors is primarily involved in the action of this drug?
- 5-HT_{1B/1D}
 - 5-HT_{2A}
 - 5-HT₃
 - 5-HT₄
- Q71. A drug administered intravenously shows a constant rate of elimination over time, irrespective of its plasma concentration. This means that a fixed amount of the drug is eliminated per unit time rather than a constant proportion. Such kinetics are typically seen when the metabolic pathways responsible for drug elimination become saturated. As a result, the rate of drug removal remains constant even with increasing drug concentration. This has important clinical implications, as small increases in dose can lead to disproportionate increases in plasma levels. Which of the following types of kinetics does this drug follow?
- First-order kinetics
 - Zero-order kinetics
 - Mixed-order kinetics
 - Michaelis-Menten kinetics
- Q72. A drug with a narrow therapeutic index requires close monitoring because the margin between its effective dose and toxic dose is very small. Even slight variations in dosage or plasma levels can lead to toxicity or therapeutic failure. Clinicians often measure this parameter to assess the safety profile of a drug. It provides an estimate of how safely a drug can be administered without causing harmful effects. Which of the following best describes the therapeutic index of a drug?
- Ratio of toxic dose to effective dose



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- b) Ratio of effective dose to toxic dose
- c) Plasma concentration of drug
- d) Rate of drug elimination

Q73. A patient develops an immediate wheal and flare reaction within minutes of exposure to an allergen. This rapid response is characteristic of a type I hypersensitivity reaction. On further evaluation, mast cell degranulation is observed with release of histamine and other inflammatory mediators. This process is triggered by cross-linking of specific immunoglobulins bound to mast cell surfaces. Which of the following immunoglobulins is primarily responsible for mediating this reaction?

- a) IgG
- b) IgA
- c) IgM
- d) IgE

Q74. A patient develops fever, rash, and joint pain approximately 7–10 days after receiving an antitoxin. These symptoms are suggestive of serum sickness, which occurs due to formation of antigen–antibody immune complexes. These complexes circulate in the bloodstream and deposit in various tissues, leading to activation of complement and inflammation. The delayed onset reflects the time required for immune complex formation. Which type of hypersensitivity reaction is involved in this condition?

- a) Type I
- b) Type II
- c) Type III
- d) Type IV

Q75. An HIV-positive patient presents with headache, fever, and features suggestive of meningitis. Cerebrospinal fluid examination reveals yeast-like organisms. On India ink preparation, the organisms appear as budding yeast surrounded by a clear halo, indicating the presence of a polysaccharide capsule. This encapsulated organism is a common opportunistic pathogen in immunocompromised individuals, particularly those with AIDS. Which of the following organisms is most likely responsible for this infection?

- a) *Candida albicans*
- b) *Aspergillus fumigatus*
- c) *Cryptococcus neoformans*
- d) *Histoplasma capsulatum*

Q76. A patient presents with white curd-like patches in the oral cavity that can be scraped off, suggestive of oral thrush. Microscopic examination using KOH mount reveals budding yeast cells with pseudohyphae. Further testing shows a positive germ tube test, which is a characteristic diagnostic feature of a specific fungal organism. This organism is a common opportunistic pathogen, especially in immunocompromised individuals or those on prolonged antibiotic therapy. Which of the following organisms is most likely responsible for this infection?

- a) *Candida albicans*
- b) *Aspergillus*
- c) *Mucor*
- d) *Cryptococcus*

Q77. A patient presents with malabsorption, steatorrhea, and foul-smelling diarrhea following consumption of untreated water. He also complains of abdominal bloating and weight loss. Stool microscopy reveals pear-shaped trophozoites with two nuclei, giving a characteristic “face-like” appearance. This organism adheres to the intestinal mucosa and interferes with fat absorption, leading to greasy stools. The infection is commonly



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associated with contaminated water sources. Which of the following parasites is most likely responsible for this condition?

- a) *Entamoeba histolytica*
- b) *Giardia lamblia*
- c) *Cryptosporidium parvum*
- d) *Cyclospora*

Q78. A patient presents with features of anemia along with marked eosinophilia on laboratory evaluation. He also complains of abdominal discomfort and nutritional deficiencies. Stool examination reveals ova that are thick-shelled with a characteristic mammillated outer layer. These findings are typical of a large intestinal roundworm infection. The parasite is known to cause malnutrition and may lead to intestinal obstruction in heavy infestations. Which of the following organisms is most likely responsible for this condition?

- a) *Trichuris trichiura*
- b) *Ascaris lumbricoides*
- c) Hookworm
- d) *Enterobius*

Q79. A 5-year-old child presents with fever followed by a generalized vesicular rash appearing in different stages, including macules, papules, vesicles, and crusts. The lesions are distributed over the trunk and face, with new lesions appearing over several days. This pattern is characteristic of a viral exanthem seen in childhood. The causative virus is a DNA virus that can remain latent in the dorsal root ganglia and may reactivate later in life. Which of the following viruses is most likely responsible for this condition?

- a) Herpes simplex virus
- b) Varicella-zoster virus
- c) Measles virus
- d) Rubella virus

Q80. A patient develops jaundice following a recent blood transfusion, raising suspicion of acute viral hepatitis. Serological testing reveals HBsAg positivity along with anti-HBc IgM, indicating an acute hepatitis B infection. Additionally, the presence of HBeAg suggests active viral replication within hepatocytes. This marker is associated with increased transmission risk and greater infectivity of the virus. What does the presence of HBeAg indicate in this patient?

- a) Past infection
- b) Immunity
- c) High infectivity
- d) Chronic inactive carrier state

Q81. A hospitalized patient receiving prolonged broad-spectrum antibiotic therapy develops profuse watery diarrhea. He may also have abdominal cramps and signs of colitis. Stool analysis reveals toxins that inactivate Rho GTP-binding proteins, leading to disruption of the actin cytoskeleton and mucosal damage. This organism is a common cause of antibiotic-associated colitis due to alteration of normal gut flora. Which of the following organisms is most likely responsible for this condition?

- a) *Clostridium perfringens*
- b) *Clostridium difficile*
- c) *Vibrio cholerae*
- d) *Salmonella*

Q82. A 30-year-old woman presents with dysuria, urinary urgency, and increased frequency of micturition, suggestive of a urinary tract infection. Urine culture reveals Gram-negative rods that ferment lactose,



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producing pink colonies on MacConkey agar. Further biochemical testing shows the organism is indole positive. These characteristics are typical of the most common causative agent of community-acquired UTIs. Which of the following organisms is most likely responsible for this infection?

- a) *Klebsiella pneumoniae*
- b) *Proteus mirabilis*
- c) *Escherichia coli*
- d) *Pseudomonas aeruginosa*

Q83. A 28-year-old patient develops acute onset vomiting and diarrhea after consuming improperly stored food. Symptoms occur rapidly, even after the food was reheated before consumption. This suggests that the toxin responsible is resistant to heat and remains active despite cooking. Such toxins are preformed in food and cause illness without requiring bacterial proliferation in the host. Which of the following types of toxins is most likely responsible for this condition?

- a) Endotoxin
- b) Heat-labile exotoxin
- c) Heat-stable exotoxin
- d) Superantigen

Q84. A microbiologist studies a bacterial strain that rapidly acquires resistance to multiple antibiotics after interaction with another bacterium. Genetic analysis reveals transfer of plasmid DNA carrying resistance genes through direct cell-to-cell contact. This process involves formation of a conjugation pilus that facilitates transfer of genetic material between bacteria. It is a major mechanism for spread of antibiotic resistance among bacterial populations. Which of the following genetic mechanisms is responsible for this phenomenon?

- a) Transformation
- b) Transduction
- c) Conjugation
- d) Mutation

Q85. A forensic expert examines a gunshot injury in a victim where the entry wound is irregular and appears elongated rather than circular. The margins are uneven, and the bullet trajectory does not appear to follow a straight path. It is suspected that the bullet lost its stability before striking the target, leading to an abnormal impact pattern. Which type of bullet is most likely responsible for producing such a keyhole-shaped entry wound?

- a) Ricochet bullet
- b) Yawing bullet
- c) Tumbling bullet
- d) Tandem bullet

Q86. A patient is brought to the emergency department following a head injury. Initially, he loses consciousness but regains it after a short period and appears relatively normal. After some time, he again deteriorates and becomes unconscious. On examination, there are signs of raised intracranial pressure. This pattern of transient recovery followed by rapid neurological decline is noted. What is the most probable diagnosis in this case?

- a) Subdural haemorrhage
- b) Extradural haemorrhage
- c) Subarachnoid haemorrhage
- d) Intracerebral haemorrhage



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- Q87. A researcher is studying the molecular structure and synthesis of collagen, a major structural protein in connective tissues. He reviews the arrangement of amino acids in collagen chains, the role of post-translational modifications, and the final assembly of collagen fibers. While analyzing structural details, he identifies one incorrect statement among commonly known features of collagen. Which of the following statements regarding collagen is incorrect?
- Triple helical structure is a right-handed superhelix
 - Glycine residue is present at every third position of α chains
 - Vitamin C is required for post-translational modification
 - Tropocollagen molecules finally attain a left-handed super-super coil
- Q88. A forensic expert is examining skeletal remains and is asked to determine the racial characteristics based on skull morphology. On assessment, the skull appears broad with a reduced anteroposterior diameter and increased transverse diameter, giving it a rounded appearance. The cephalic index is calculated and found to be high, consistent with a brachycephalic skull type. This type of skull is most commonly seen in which of the following populations?
- Mongolians
 - Europeans
 - South Indians
 - Negroes
- Q89. A convict escapes from prison, and no biological samples or family reference records are available with the authorities. A mutilated dead body resembling the convict is later discovered in a nearby forest. Due to severe facial disfigurement, visual identification is not possible. Since no comparative biological samples are available, investigators must rely on alternative scientific methods for establishing identity. Which of the following methods can provide positive identification in this case?
- Blood grouping
 - DNA profiling
 - Anthropometry
 - HLA typing
- Q90. A dead body is brought for medico-legal examination with a ligature mark present around the neck. As per police history, the body was found in a partially suspended position with the legs touching the ground. There are no signs of struggle, and the findings suggest partial hanging. It is known that even a small amount of body weight can cause fatal compression of vital neck structures. What is the approximate minimum weight required to cause death in such cases?
- 1.15 kg
 - 2.20 kg
 - 3.10 kg
 - 4.50 kg
- Q91. A decomposed body is brought for autopsy with suspicion of death due to prolonged starvation. On internal examination, characteristic features of nutritional deprivation are assessed. The pathologist carefully evaluates the gastrointestinal tract and associated organs for classical findings seen in starvation. However, one of the listed findings is inconsistent with death due to starvation. Which of the following is NOT seen in death due to starvation?
- Translucent intestinal walls
 - Contracted and empty stomach
 - Large bowel may contain faecolith



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d) Shrunken gall bladder

Q92. A patient is brought to the emergency department with altered sensorium, metabolic acidosis, and visual disturbances. History reveals ingestion of a toxic alcohol. Laboratory findings show a high anion gap metabolic acidosis with elevated osmolar gap. The treating physician decides to administer a specific antidote that inhibits alcohol dehydrogenase to prevent formation of toxic metabolites. Fomepizole is indicated in the poisoning of which of the following substances?

- a) Organophosphorus compounds
- b) Methyl alcohol
- c) Aluminium phosphide
- d) Copper

Q93. A medico-legal case is being evaluated where a male is accused of impotence. On detailed assessment, it is found that the individual is capable of performing sexual intercourse with some partners but fails to do so with a particular woman. There is no structural or physiological abnormality detected. Which type of impotence best describes this condition?

- a) Inability to beget children or conceive
- b) Impotent quoad hanc (relative impotence)
- c) Inability to perform sexual intercourse
- d) Inability to initiate or maintain sexual arousal in female

Q94. A medical graduate has obtained his qualification from a foreign institution outside India. He wishes to get his qualification recognized and registered under the provisions of the Indian Medical Council Act, 1956. The eligibility depends on whether the qualification is included in the appropriate schedule of the Act. In which schedule are medical qualifications awarded by institutions outside India and recognized by MCI listed?

- a) First Schedule of Indian Medical Council Act, 1956
- b) Second Schedule
- c) Part I of Third Schedule
- d) Part II of Third Schedule

Q95. All are correct about given image except:



- a) Step approach phenomenon
- b) Surveillance of both communicable and non-communicable diseases
- c) Surveillance of non-communicable diseases only
- d) Personal, Physical, and Biochemical assessment

Q96. Team used to describe a study which is longitudinal, observational, started with exposed and nonexposed group and includes analysis of variables to determine association:

- a) Case control study
- b) Prospective cohort study



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- c) Retrospective cohort study
- d) RCT

Q97. Incorrect match about malaria diagnosis:



- a) Slide preparation- MPW-M
- b) RDT Kit test- ASHA
- c) Slide checking and reporting- MPW-M
- d) Best test for malaria diagnosis- thick and thin blood smear

Q98. All are considered high risk groups for given program except:



- a) I/V drug addicts
- b) Sex workers
- c) Long distance truck drivers
- d) Men having sex with men (MSM)

Q99. Tuberculin test is cheap, easily available and performed but losing its value for detection of latent TB infection due to:



- a) Low incidence of disease in community
- b) High incidence of disease in community
- c) Low BCG vaccine coverage of community
- d) High BCG vaccine coverage of community



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Q100. As per national immunisation schedule under universal immunization program, vaccine given at 9 month by intramuscular route in left thigh is:

- a) Edmonston jagreb
- b) P1,2,3
- c) SA 14-14-2
- d) Kolar strain vaccine

Q101. 4DS of Rashtriya Bal Swasthya Karyakram includes :

RBSK

RASHTRIYA BAL SWASTHYA KARYAKRAM
राष्ट्रीय बाल स्वास्थ्य कार्यक्रम
FROM SURVIVAL TO HEALTHY SURVIVAL

- a) Diseases, Deficiency, Dandruff, Diarrhoea
- b) Diseases, Deficiency, Deformities, Developmental delay
- c) Deafness, Deficiency, Deformities, Developmental delay
- d) Dehydration, Deficiency, dementia, Diarrhoea

Q102. After birth of desired last child, family will enter in which phase?

- a) Extension
- b) Complete extension
- c) Contraction
- d) Complete contraction

Q103. All are approaches of health education of community except:

- a) Regulatory approach
- b) Service approach
- c) Health education approach
- d) Appropriate technology

Q104. Phase of demographic cycle where demographic gap starts increasing in beginning and reaches to maximum towards end:

- a) High stationary
- b) Low stationary
- c) Early expansion
- d) Late expansion

Q105. 25year female XXX has delivered a baby by operation (Cesarean section). She has already given consent for Cu IUCD insertion for birth spacing, time for insertion of this device:

- a) 24 hours after delivery by vaginal route
- b) After delivery of baby and abdominal closure by vaginal route
- c) After delivery of baby and placenta but before uterus and abdominal wall closure
- d) Doctor will refuse for insertion as not safe after operation

Q106. Incorrect about ASHA worker:

- a) 600Rs as incentive for motivating hospital delivery in rural high performance states
- b) Distribute anti TB medication



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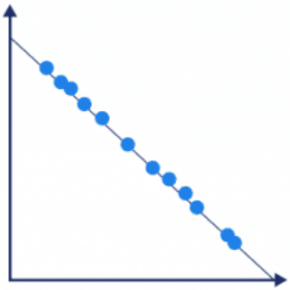
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- c) Population coverage- 1/1000
- d) Conduction of delivery

Q107. Incorrect match about concern worker:

- a) Td vaccine to pregnant patient in village-ASHA
- b) Growth monitoring- Anganwadi
- c) Well chlorination – MPW-M
- d) MCH care- MPW-F

Q108. A study was conducted to find association between exercise duration and blood sugar level and observed data was plotted as per given image. CALCULATE correlation coefficient of given situation:



- a) 0
- b) +1
- c) -1
- d) -1 to +1

Q109. Deviation of a value from mean value in terms of standard deviation is known as:

- a) Range
- b) Standard deviation
- c) Relative deviate
- d) Confidence level

Q110. A 6-year-old child undergoes tonsillectomy for recurrent tonsillitis and has an uneventful immediate postoperative recovery. Later in the evening, the nursing staff observes active bleeding from the mouth, which appears profuse. The child becomes restless and is at risk of aspiration. On examination, there is ongoing oropharyngeal bleeding. What is the most appropriate next step in management?

- a) Reassure and discharge as mild bleeding is expected
- b) Admit and start antibiotics
- c) Immediate re-exploration in operation theatre
- d) Start oral hematinics

Q111. A patient presents to the emergency department following head trauma with persistent clear nasal discharge. The fluid increases on bending forward and is suspected to be cerebrospinal fluid. A sample is placed on filter paper, which shows a characteristic “double halo” or “ring” sign, with a central blood spot surrounded by a clear peripheral ring. There are associated features of skull base injury. What is this finding most indicative of?

- a) CSF rhinorrhoea
- b) Traumatic CSF leak



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- c) Fracture mastoid
- d) Meningoencephalocele

Q112. A 7-year-old child presents after sustaining nasal trauma while playing. He complains of nasal obstruction and pain. On examination, there is swelling over the nose with bilateral soft, fluctuant swelling of the nasal septum causing obstruction of both nasal cavities. The swelling is tender and reddish in appearance. There is no active external bleeding. What is the most appropriate next step in management?

- a) Incision and drainage of septal swelling
- b) Anterior nasal packing
- c) Start intravenous steroids
- d) Start topical corticosteroid nasal spray

Q113. A critically ill patient is admitted to the ICU and requires prolonged ventilatory support. Due to anticipated long-term airway management, a tracheostomy is planned. The patient is on mechanical ventilation and requires protection against aspiration. Continuous monitoring and airway security are essential in this setting. Which type of tracheostomy tube is most appropriate for this patient?

- a) Metallic tracheostomy tube
- b) PVC cuffed tracheostomy tube
- c) PVC uncuffed tracheostomy tube
- d) Fenestrated uncuffed tracheostomy tube

Q114. A patient with known T3 laryngeal carcinoma presents to the emergency department with severe respiratory distress. He has noisy breathing with inspiratory stridor and is unable to maintain adequate oxygenation. There is evidence of upper airway obstruction due to the tumor mass. The patient appears anxious and tachypneic, indicating impending airway compromise. Immediate intervention is required to secure the airway. What is the most appropriate immediate management in this patient?

- a) Total laryngectomy with postoperative radiotherapy
- b) Radiotherapy
- c) Laser cordectomy
- d) Tracheostomy

Q115. A medical student is being questioned about the anatomical relations of the anterior (carotid) wall of the middle ear during a practical examination. He recalls that this wall contains structures that communicate with the nasopharynx and transmit important muscles. However, he is unsure which of the listed structures is not directly related to this wall. Which of the following structures is NOT related to the anterior wall of the middle ear?

- a) Eustachian tube
- b) Canal for tensor tympani
- c) Peritubal air cells
- d) Processus cochleariformis

Q116. A 16-year-old boy presents with complaints of pain over the cheek region associated with nasal obstruction and purulent nasal discharge. The symptoms have been progressively worsening over the past few days. On anterior rhinoscopy, thick purulent discharge is noted in the middle meatus. There is localized tenderness over the cheek on palpation. Which of the following paranasal sinuses is most likely involved in this case?

- a) Maxillary sinus
- b) Ethmoid sinus
- c) Frontal sinus



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d) Sphenoid sinus

Q117. A newborn presents immediately after birth with respiratory distress, cyanosis, and difficulty in breathing. The symptoms worsen during feeding and improve when the baby cries. On examination, bilateral nasal obstruction is suspected, and attempts to pass a nasal catheter fail. A diagnosis of bilateral choanal atresia is considered. What is the primary reason this condition becomes a life-threatening airway emergency in neonates?

- a) Associated pulmonary agenesis
- b) Associated pulmonary hypoplasia
- c) Neonates are obligatory nasal breathers
- d) Associated mandibular hypoplasia

Q118. A 45-year-old woman was admitted to the ICU following a head injury and required mechanical ventilation with endotracheal intubation for 5 days. After recovery, she is shifted to the ward and subsequently complains of persistent hoarseness of voice. There is no history of prior voice disorder or smoking. Indirect laryngoscopy reveals a lesion over the posterior part of the vocal cords. What is the most likely etiology of her hoarseness?

- a) Pneumonia
- b) Vocal polyp
- c) Intubation granuloma
- d) Vocal nodule

Q119. A 7-year-old child undergoes tonsillectomy for recurrent tonsillitis and has an uneventful immediate postoperative recovery. Later in the evening, the nursing staff notices profuse bleeding from the child's mouth. The child becomes anxious and restless, with ongoing active oropharyngeal bleeding. There is a risk of aspiration and hemodynamic instability if not managed promptly. What is the most appropriate next step in management?

- a) Reassure and discharge as mild bleeding is expected
- b) Admit and start antibiotics
- c) Immediate re-exploration in operation theatre
- d) Start oral hematinics

Q120. A 7-year-old girl is brought to the outpatient clinic with complaints of difficulty seeing the blackboard at school. She was recently prescribed corrective spectacles after refraction testing. However, on follow-up examination, the visual acuity in the right eye does not improve beyond 6/18 despite appropriate glasses. There is no history of trauma, redness, or pain in the eye. Fundus examination is normal. What is the most appropriate next step in management?

- a) LASIK
- b) Observation
- c) Occlusion therapy
- d) Contact lenses

Q121. A patient who previously underwent corneal transplantation presents with complaints of decreased vision, redness, and mild ocular discomfort. On slit-lamp examination, there is corneal edema with keratic precipitates. A distinct linear opacity is observed on the corneal endothelium at the graft-host junction, progressing centrally. This finding is characteristic of endothelial rejection of the corneal graft. What is this clinical sign called?

- a) Vogt's line



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- b) Arlt's triangle
- c) Stocker's line
- d) Khodadoust line

Q122. A middle-aged man is diagnosed with cataract in the left eye and is scheduled for phacoemulsification with intraocular lens (IOL) implantation. He has a past history of LASIK surgery performed 10 years ago for correction of high myopia. Preoperative evaluation is carried out to determine the appropriate IOL power. However, the surgeon anticipates difficulty in accurate calculation due to previous refractive surgery. Which of the following factors most significantly affects the accuracy of IOL power calculation in this patient?

- a) Axial length of the eyeball
- b) Changed curvature of the lens
- c) Changed keratometry
- d) Fundus changes due to high myopia

Q123. A young patient presents to the emergency department after being struck in the eye by a tennis ball during play. He complains of double vision and pain around the orbit. On examination, there is enophthalmos and restricted upward gaze. Sensory examination reveals decreased sensation over the lower eyelid and cheek. Fundus examination is unremarkable. What is the most likely diagnosis in this patient?

- a) Macular edema
- b) Uveitis
- c) Blow-out fracture
- d) Rosette cataract

Q124. A 60-year-old woman with a known history of rheumatoid arthritis presents with severe ocular pain and progressive diminution of vision. She also complains of photophobia and redness of the affected eye. On examination, there is diffuse congestion with a characteristic deep violaceous (violet-red) hue seen in natural light. Visual acuity is reduced to 6/60. The pain is severe and radiates around the orbit. What is the most likely diagnosis in this patient?

- a) Dry eyes
- b) Episcleritis
- c) Scleritis
- d) Anterior uveitis

Q125. A 65-year-old patient presents to the ophthalmology clinic 3 days after undergoing uneventful cataract surgery. He reports increasing ocular pain and rapidly progressive diminution of vision after an initial period of improvement. There is associated redness and photophobia. On examination, the eye appears inflamed with reduced visual acuity. What is the most likely cause of this presentation?

- a) Vitreous haemorrhage
- b) Endophthalmitis
- c) Secondary cataract
- d) Retinal detachment

Q126. An in-hospital workup of a 78-year-old hypertensive, mildly asthmatic man who is receiving chemotherapy for colon cancer reveals symptomatic gallstones. Preoperative laboratory results are notable for a hematocrit of 24% and urinalysis with 18 to 25 WBCs and gram-negative bacteria. On call to the operating room, the patient receives intravenous penicillin. His abdomen is shaved in the operating room. An open cholecystectomy is performed and, despite a lack of indications, the common bile duct is explored. The wound is closed primarily with a Penrose drain exiting a separate stab wound. On postoperative day 3, the



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patient develops a wound infection. Which of the following changes in the care of this patient could have decreased the chance of a postoperative wound infection?

- a) Increasing the length of the preoperative hospital stay to prophylactically treat the asthma with steroids
- b) Treating the urinary infection prior to surgery
- c) Shaving the abdomen the night prior to surgery
- d) Using a closed drainage system brought out through the operative incision

Q127. A 39-year-old previously healthy male is hospitalized for 2 weeks with epigastric pain radiating to his back, nausea, and vomiting. Initial laboratory values revealed an elevated amylase level consistent with acute pancreatitis. Five weeks following discharge, he complains of early satiety, epigastric pain, and fevers. On presentation, his temperature is 38.9°C (102°F) and his heart rate is 120 beats per minute; his white blood cell (WBC) count is 24,000/mm³ and his amylase level is normal. He undergoes a CT scan demonstrating a 6 cm by 6 cm rim-enhancing fluid collection in the body of the pancreas. Which of the following would be the most definitive management of the fluid collection?

- a) Antibiotic therapy alone
- b) Antibiotics and CT-guided aspiration with repeat imaging in 2 to 3 days
- c) Antibiotics and percutaneous catheter drainage
- d) Surgical internal drainage of the fluid collection with a cyst-gastrostomy or Roux-en-Y cyst- jejunostomy

Q128. Mrs. X 55 years old underwent a surgical procedure in the neck for a multinodular goiter involving both the lobes of Thyroid gland for her pressure symptoms of Dyspnoea and Dysphagia. There was a risk of developing Follicular thyroid cancer as well. Her operation was uneventful and was shifted to the recovery room with stable vitals. Suddenly approximately 6 hours of operation the patient started coughing and choking sensation with severe difficulty in breathing. Immediately you who was the resident doctor on duty, opened up her skin sutures in the neck and saved your patient. What can be the most likely cause of her breathing difficulty in this situation?

- a) Damage to both the Recurrent laryngeal nerves during the operation by the surgeon
- b) Damage to the External branch of Superior laryngeal nerve on right side during the operation by the surgeon
- c) Reactionary hemorrhage due to slipping of ligature from the Superior thyroid artery in the postoperative period
- d) Secondary hemorrhage due to infection eroding the wall of the blood vessel in the postoperative period

Q129. A 35-year-old woman presents with a painless lump in the upper inner quadrant of her left breast. On examination, the lump measures approximately 6 cm, is soft to firm in consistency, well-circumscribed, and freely mobile over the pectoralis major muscle. The overlying skin appears normal, and there is no axillary lymphadenopathy. Triple assessment, including ultrasound-guided biopsy, suggests a benign lesion. She is planned for elective surgical excision of the lump under aseptic conditions. Under which category does this surgical procedure fall?

- a) Clean
- b) Clean-contaminated
- c) Contaminated
- d) Dirty

Q130. A 34-year-old woman presents with a painless lump in her left breast, which she noticed recently during self-examination. The lump is approximately the size of a marble, well-defined, and mobile, without fixation to underlying structures. There is no associated nipple discharge or skin changes. On clinical examination, axillary lymph nodes are not palpable. She gives a significant family history of breast carcinoma, with her mother having died from the disease. What is the most appropriate next step in management?



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- a) Mammography
- b) Ultrasound of the breast
- c) Fine needle biopsy
- d) Excision biopsy of the lump

Q131. A 45-year-old patient undergoes biopsy for a suspected neoplasm, and histopathological evaluation is performed. The pathology report describes the tumor as Grade I on a grading scale ranging from I to IV. The report further comments on cellular morphology, architectural pattern, and resemblance to the tissue of origin. There is minimal nuclear atypia and low mitotic activity noted. Which of the following descriptions best explains the basis of this grading?

- a) It cannot have an in-situ component
- b) It has arisen only from epithelium
- c) It is unlikely to be malignant
- d) It is well differentiated

Q132. A 60-year-old man presents with painless hematuria and is evaluated by a urologist. Urine cytology reveals malignant cells, raising suspicion of urothelial carcinoma. Cystoscopic examination demonstrates a superficial papillary lesion confined to the mucosa and submucosa. Histopathology confirms transitional cell carcinoma without evidence of muscle invasion (Stage A). The patient is otherwise stable, with no signs of metastatic disease. What is the most appropriate management for this condition?

- a) Topical (intravesical) chemotherapy
- b) Radical cystectomy
- c) Radiation therapy
- d) Transurethral resection of bladder tumor with intravesical chemotherapy

Q133. A 45-year-old man presents with iron deficiency anemia requiring repeated transfusions and is evaluated with colonoscopy. A mass lesion is identified at the hepatic flexure of the colon, and biopsy confirms adenocarcinoma. During exploratory laparotomy, a solitary 3-cm metastatic lesion is incidentally found at the edge of the right lobe of the liver. There is no evidence of additional metastatic disease, and the patient was preoperatively counseled regarding such findings. He remains hemodynamically stable throughout the procedure. What is the most appropriate management in this case?

- a) Perform diverting ileostomy and obtain further imaging
- b) Right hemicolectomy alone
- c) Right hemicolectomy with local resection of the liver metastasis
- d) Close abdomen and initiate chemotherapy

Q134. A 22-year-old man sustains a penetrating abdominal injury due to a gunshot wound and undergoes exploratory laparotomy. A solitary distal small bowel injury is identified and managed with resection and primary anastomosis. On postoperative day 7, he develops leakage of small bowel contents through the surgical incision. The fascial layer remains intact, and there are no signs of intra-abdominal sepsis. The fistula output is approximately 300 mL/day. What is the most appropriate management in this patient?

- a) Early reoperation to close the fistula tract
- b) Broad-spectrum antibiotics
- c) Total parenteral nutrition
- d) Somatostatin to reduce fistula output

Q135. A 60yr old male after heavy meal went for a walk. He developed severe chest pain associated with vomiting. He was brought to emergency room and contrast swallow study showed the following picture, most probable diagnosis in this patient is :



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- a) Acute MI with myocardial rupture
- b) Rupture of emphysematous bullae
- c) Boerhaave's syndrome
- d) Mallory weiss syndrome

Q136. A 30-year-old man is brought to the emergency department following a road traffic accident. On arrival, he is restless and irritable. He opens his eyes only in response to painful stimulus. His verbal response consists of incomprehensible sounds. On motor assessment, he demonstrates abnormal flexion of the upper limbs to pain. Peripheral pulses are absent, although the carotid pulse is still palpable, suggesting severe shock. What is the correct Glasgow Coma Scale (GCS) score in this patient?

- a) E2, V3, M4
- b) E2, VNT, M5
- c) E2, V2, M3
- d) E3, V3, M4

Q137. What should be the treatment done after needle thoracocentesis and stabilisation, in a patient who presented in the emergency with respiratory distress. He was having dilated neck veins at presentation. Large amount of air was seen on the left side along with mediastinal shift as shown in the image taken at arrival in ER?



- a) Wide bore needle insertion in the intercostal space
- b) Endotracheal intubation
- c) Tracheostomy
- d) Tube thoracocentesis and breathing exercises

Q138. A 2-year-old child is brought to the ER with severe abdominal pain. On performing emergency exploratory laparotomy, the following finding is noted. Which of the following statements is false regarding this condition?

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- a) Cannot be a lead point for intussusception
- b) The most common presenting symptom is bleeding in younger children
- c) It is a true diverticulum which extends from the anti-mesenteric side
- d) Most common heterotopic tissue is gastric mucosa

Q139. 72 year old male was brought to the emergency in a state of shock with no peripheral pulses palpable, systolic blood pressure unrecordable in the upper limbs, cool clammy peripheries. There is no history of any kind of trauma. Basic resuscitation with IV Fluids was started and blood requested from the blood bank. A high suspicion of huge amount of blood in the retroperitoneum is there with very high chance of mortality in this patient. Which of the following cause is least explanatory of the given situation ?

- a) Rupture of pseudocyst around the pancreas
- b) Major bleed from the rupture of hepatic hemangioma
- c) Splenic artery aneurysm rupture in a patient of portal hypertension
- d) Rupture of abdominal aortic aneurysm

Q140. Identify the instrument in the image given below:



- a) Circular stapler
- b) Linear stapler
- c) Curvilinear stapler
- d) Skin stapler

Q141. A 35-year-old woman presents with infertility following prior radiation therapy for Hodgkin's lymphoma. Radiation can damage ovarian follicles, leading to diminished ovarian reserve. To evaluate her reproductive potential, assessment of ovarian reserve is essential. Among available tests, one marker directly reflects the number of remaining primordial follicles and is relatively cycle-independent. Which of the following is the best indicator of ovarian reserve in this patient?

- a) LH level
- b) LH/FSH ratio
- c) FSH level
- d) Anti-Müllerian hormone levels



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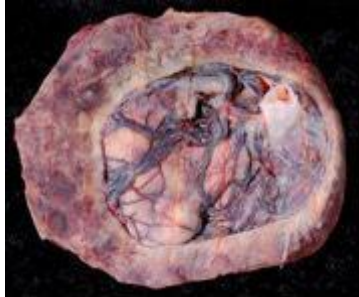
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- Q142. A nursing assistant in the obstetric unit is monitoring labour using a partograph. The cervical dilatation is plotted against time to assess the progress of labour. The alert line represents expected progress, while deviation from this line indicates abnormal labour progression. When the plotting of cervical dilatation moves to the right of the alert line, it suggests slow or prolonged labour and the need for closer monitoring or referral. Further delay crossing the action line indicates need for intervention. Where would cervical dilatation marking indicate prolonged labour?
- a) Right of alert line
 - b) Left of alert line
 - c) Right of action line
 - d) Anywhere
- Q143. A 34-year-old pregnant woman at 30 weeks gestation presents with acute onset abdominal pain, vaginal bleeding, and absence of fetal movements. On examination, the abdomen is tense, hard, and tender, indicating uterine irritability. Fetal heart sounds are not audible, suggesting fetal compromise or demise. This clinical picture is characteristic of a condition where the placenta prematurely separates from the uterine wall, leading to concealed or revealed hemorrhage and uterine rigidity. What is the most likely diagnosis in this patient?
- a) Polyhydramnios
 - b) Severe oligohydramnios
 - c) Abruptio placentae
 - d) Placenta previa
- Q144. A 33-year-old woman is brought to the emergency in a state of shock. History reveals amenorrhea along with a previously positive pregnancy test. She may also have abdominal pain and vaginal bleeding. In early pregnancy, the combination of shock with positive pregnancy strongly suggests internal hemorrhage. The most common life-threatening cause in this scenario is rupture of an ectopic pregnancy, leading to massive intra-abdominal bleeding. What is the most likely diagnosis in this patient?
- a) Ruptured ectopic pregnancy
 - b) Molar pregnancy
 - c) Preterm labour
 - d) Postpartum hemorrhage
- Q145. A woman who delivered 6 months ago presents with persistent irregular vaginal bleeding. She is hemodynamically stable but continues to have bleeding despite undergoing dilatation and curettage 3 months earlier. Normal postpartum bleeding (lochia) should resolve within a few weeks, and persistent bleeding months later is abnormal. The history raises suspicion of a gestational trophoblastic neoplasm, especially when bleeding persists after uterine evacuation. Choriocarcinoma is a highly vascular malignant condition that commonly presents with irregular bleeding following pregnancy. What is the most likely diagnosis in this patient?
- a) Postpartum hemorrhage
 - b) Lochia
 - c) Disseminated intravascular coagulation
 - d) Choriocarcinoma
- Q146. A primigravida delivered a baby at term with a birth weight of 1.9 kg via normal vaginal delivery. The placenta delivered appears as shown in the image below. What is this condition known as?



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- a) Placenta succenturiata
- b) Battledore placenta
- c) Circumvallate placenta
- d) Bipartite placenta

Q147. A 25-year-old primigravida has delivered a healthy term baby via normal vaginal delivery. She is currently in the early postnatal period and is clinically stable. During discharge counseling, she expresses concern regarding persistent vaginal discharge since delivery. She reports that the discharge is initially reddish and has gradually decreased in amount. There is no associated foul odor, fever, or abdominal pain. On examination, uterine involution is appropriate for the postpartum day. What is the most appropriate advice regarding her vaginal discharge?

- a) The discharge is abnormal and requires immediate evaluation
- b) The amount of discharge directly correlates with lactation
- c) The discharge will persist for at least 2 weeks
- d) The color changes from bright red to yellow to brown

Q148. A multiparous woman in active labor presents with prolonged labor and a history of previous lower segment cesarean section performed two years ago. During labor, she suddenly develops severe abdominal pain followed by vaginal bleeding and signs of hemodynamic instability. On examination, her blood pressure is falling and pulse rate is increased. Per abdominal examination reveals easily palpable fetal parts with loss of uterine contour. Fetal heart sounds are absent. Ultrasonography demonstrates intrauterine fetal demise along with free fluid in the peritoneal cavity. What is the most likely diagnosis?

- a) Uterine dehiscence
- b) Obstructed labor
- c) Abruptio placentae
- d) Uterine rupture

Q149. A postpartum woman with a recent history of eclampsia is receiving intravenous magnesium sulphate infusion as part of standard management. After a few hours, she becomes progressively drowsy and complains of generalized weakness with excessive sweating. On clinical examination, her deep tendon reflexes are absent, particularly bilateral knee jerks. Her respiratory effort appears reduced, although blood pressure remains stable. There is no evidence of ongoing seizures or focal neurological deficit. What is the most likely cause of her current clinical condition?

- a) HELLP syndrome
- b) Worsening of eclampsia
- c) Magnesium sulphate toxicity
- d) Disseminated intravascular coagulation

Q150. A 30-year-old primigravida at 32 weeks of gestation presents for a routine antenatal evaluation. She reports no significant complaints and has had an uneventful pregnancy so far. On abdominal examination, the uterine size is noted to be smaller than expected, corresponding to approximately 28 weeks of gestation. Fetal



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movements are perceived but appear reduced. Ultrasound examination reveals a markedly decreased amniotic fluid index (AFI) of 4 cm. No obvious placental abnormalities are noted. Which of the following fetal anomalies is most likely associated with this finding?

- a) Renal agenesis
- b) Neural tube defect
- c) Oesophageal atresia
- d) Cleft palate

Q151. A full-term newborn presents with central cyanosis soon after birth and requires oxygen support. Clinical evaluation reveals no significant respiratory distress, but oxygen saturation remains low despite adequate ventilation. Echocardiography confirms the diagnosis of transposition of the great arteries. The baby is stabilized and further evaluation is planned. Detailed maternal history is sought to identify potential risk factors associated with this congenital anomaly. Which of the following maternal conditions should be specifically investigated in this case?

- a) Maternal preeclampsia
- b) Maternal atrial septal defect
- c) Diabetes mellitus
- d) Maternal transposition of great arteries

Q152. A 32-year-old woman presents with progressively worsening dysmenorrhea and deep dyspareunia over the past two years. Her symptoms have become severe enough to interfere with daily activities. On diagnostic laparoscopy, multiple bluish-black “powder burn” deposits are noted over the peritoneum and omentum. Additionally, the right ovary reveals a cystic lesion measuring approximately 7×8 cm, suggestive of an endometrioma. There is no evidence of malignancy. What is the most appropriate management in this patient?

- a) Intraperitoneal GnRH injection
- b) Ovarian cyst aspiration with fulguration of deposits
- c) Ovarian cystectomy
- d) Ovarian cystectomy with fulguration of deposits

Q153. A multigravida woman at term presents in active labor with regular uterine contractions. She has a history of one previous lower segment cesarean section. On clinical examination, the cervix is 2 cm dilated with intact membranes, and contractions are adequate. There is no evidence of scar tenderness or fetal distress. Pelvic assessment reveals an interspinous diameter of 8 cm, suggestive of a contracted mid-pelvis. What is the most appropriate next step in management?

- a) Allow spontaneous labor with Ritgen maneuver and monitor scar tenderness
- b) Allow spontaneous labor with liberal episiotomy
- c) Forceps application in second stage
- d) Immediate LSCS

Q154. A 26-year-old primigravida at 16 weeks of gestation presents with lower abdominal pain and vaginal bleeding. The symptoms have progressively increased over the past few hours. On per-speculum examination, active bleeding is noted through the cervical os. Bimanual examination reveals a uterus corresponding to 16 weeks size, which is soft and non-tender. The cervical os is found to be open, with no expulsion of products of conception yet. What is the most appropriate diagnosis in this patient?

- a) Abruptio placentae
- b) Placenta previa
- c) Inevitable abortion
- d) Incomplete abortion



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Q155. A young sexually active woman presents with lower abdominal pain, fever, and vaginal discharge suggestive of pelvic inflammatory disease. Despite initial medical management, her symptoms persist, and diagnostic laparoscopy is performed. The procedure reveals characteristic “violin-string” adhesions between the liver capsule and the anterior abdominal wall. There is no evidence of gallbladder pathology or tubo-ovarian abscess formation. These findings are consistent with a known complication of PID involving the upper abdomen. What is the most accurate statement regarding this condition?



- a) Perihepatitis
- b) Cholecystitis
- c) Tuboovarian abscess
- d) Gonococcal urethritis

Q156. A 45-year-old patient undergoes total thyroidectomy for a follicular thyroid neoplasm. Within 24 hours post-surgery, he develops tingling sensations and neuromuscular irritability, suggestive of tetany. These symptoms are commonly due to hypocalcemia caused by inadvertent damage or removal of parathyroid glands during surgery. Early identification of hypocalcemia is crucial to prevent complications such as seizures or cardiac arrhythmias. Immediate biochemical assessment is required to guide urgent management. Which of the following serum tests should be ordered immediately?

- a) Parathormone
- b) TSH
- c) Calcium
- d) Iodine

Q157. A 55-year-old patient presents with acute onset back pain followed by progressive paresthesia and ascending weakness in both lower limbs. The symptoms began after a recent episode of diarrhea, suggesting a post-infectious etiology. On examination, there is symmetrical weakness, absent deep tendon reflexes, and flexor plantar responses, which are characteristic of Guillain–Barré syndrome. As the disease can rapidly progress to involve respiratory muscles, early assessment of respiratory function is critical. Which of the following is the most important investigation to be performed immediately in this patient?

- a) Lumbar puncture
- b) Forced vital capacity
- c) Nerve conduction study
- d) MRI brain

Q158. A 42-year-old chronic smoker and alcoholic presents with progressive shortness of breath that worsens in the supine position. Examination reveals bilateral infrascapular dullness on percussion with absent breath sounds, suggesting pleural effusion. Ultrasound shows hepatomegaly with cirrhosis, indicating underlying liver disease. In cirrhosis, pleural effusion (hepatic hydrothorax) typically occurs due to increased hydrostatic pressure and is transudative in nature. Which of the following statements is true regarding this condition?

- a) Community-acquired pneumonia
- b) Likely cause is transudative pleural effusion
- c) Likely pleural fluid protein/serum protein ratio > 0.5



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d) Good reversibility of FEV1 > 12%

Q159. A 50-year-old man with a significant 30 pack-year smoking history presents with chronic cough, progressive dyspnea, and wheezing. Examination reveals wheeze and occasional crackles, and chest X-ray shows hyperinflation, suggestive of chronic obstructive pulmonary disease (COPD). For appropriate pharmacological management, it is essential to assess the severity of airflow obstruction and its reversibility. This helps differentiate COPD from asthma and guides stepwise therapy. Which of the following is the most useful investigation to guide pharmacological management in this patient?

- a) CT chest
- b) Alpha-1 antitrypsin deficiency testing
- c) Echocardiography
- d) Spirometry with reversibility

Q160. A 70-year-old man with a history of hypertension presents with mild exertional breathlessness. On examination, he is found to have atrial fibrillation with a ventricular rate of 100/min. Echocardiography shows a dilated left atrium and mild left ventricular dysfunction. In such patients, management focuses on two key aspects: rate control and prevention of thromboembolism. Beta-blockers are preferred for rate control, especially in patients with underlying cardiac dysfunction, and anticoagulation is indicated due to increased stroke risk. Which of the following is the most appropriate initial medical management?

- a) Beta blocker and warfarin
- b) Beta blocker and aspirin
- c) Digoxin and warfarin
- d) Furosemide and warfarin

Q161. A 45-year-old man presents with acute dyspnea, fever, and cough following recent binge alcohol intake. He is tachypneic, hypotensive, and severely hypoxic with SpO₂ of 68% on room air. ABG shows hypoxemia with mild acidosis, suggesting acute respiratory failure. Chronic alcohol use with recent binge raises strong suspicion for acute pancreatitis, which can lead to systemic inflammatory response and secondary acute lung injury (ARDS). This explains severe hypoxia without primary lung pathology. Which of the following is the most likely underlying cause of this presentation?

- a) Lung contusion
- b) Pancreatitis
- c) Shock
- d) Pneumonia

Q162. A 67-year-old man with a history of hypertension presents with gradually progressive shortness of breath on exertion and orthopnea over several months. Initial investigations including blood tests, chest X-ray, and spirometry are normal, making primary pulmonary causes less likely. In suspected heart failure, a non-invasive screening test is first used to assess the likelihood of cardiac dysfunction. This test helps determine whether further imaging such as echocardiography is required. Which of the following is the most appropriate next test to perform in this patient?

- a) Troponin I
- b) B-type natriuretic peptide
- c) Myocardial perfusion scan
- d) Echocardiogram

Q163. A 45-year-old patient presents with recurrent upper respiratory symptoms and bloody nasal discharge, suggestive of chronic sinus involvement. He subsequently develops palpable purpura, indicating small-vessel vasculitis, along with hemoptysis. CT thorax reveals multiple cavitory lung lesions. The combination of upper



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airway involvement, pulmonary cavitations, and systemic vasculitic features is highly characteristic of a necrotizing granulomatous vasculitis affecting small to medium vessels. Which of the following is the most likely diagnosis?



- a) Wegener's granulomatosis
- b) Kawasaki disease
- c) Tuberculosis
- d) Cavitory metastasis

Q164. A 45-year-old woman undergoes routine occupational health screening and is found to have the following serological profile: HBsAg negative, anti-HBs negative, anti-HBc positive, HBeAg negative, and anti-HBe negative. She is asymptomatic with no clinical features of active hepatitis. This serological pattern indicates absence of surface antigen and antibodies, with persistence of core antibody. Such a profile is typically seen in a transitional phase of infection where surface antigen has disappeared but protective antibodies have not yet appeared. Which of the following best describes this woman's hepatitis B status?

- a) Persistent infection, high infectivity
- b) Persistent infection, low infectivity
- c) Pre-core mutant carrier
- d) Window period

Q165. A 29-year-old woman presents with complaints of decreased concentration and intolerance to heat, stating that her surroundings feel excessively warm. She reports increased appetite despite significant weight loss of 5 kg over the past 2 months. On examination, she is mildly febrile with tachycardia, increased respiratory rate, and elevated blood pressure. These findings suggest a hypermetabolic state. The clinical picture is highly suggestive of hyperthyroidism, where increased thyroid hormone levels lead to systemic manifestations. Which of the following laboratory findings is most likely to be present in this patient?

- a) Cortisol increase
- b) Decreased iodine uptake
- c) Increase T3 and T4
- d) Decreased plasma insulin

Q166. The child described shows developmental milestones consistent with around 18 months of age. He can walk independently, climb stairs with support, feed himself with a cup and spoon without spilling, and perform social gestures like waving bye-bye. These are typical features of a toddler in this age group. In fine motor development, an 18-month-old child is usually able to build a tower of approximately 4 cubes. Since this option is not provided, the closest correct choice among the given options is higher milestone corresponding to slightly older age. Which of the following is the most appropriate answer?

- a) 2 cubes
- b) 6 cubes
- c) 10 cubes



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d) None of these

Q167. A newborn is assessed immediately after birth for overall condition using standard clinical parameters. The infant is lying supine with flexed limbs, indicating good muscle tone. Heart rate is recorded at 126/min, which is within normal limits. On stimulation with a nasogastric tube, the baby cries actively, showing a good reflex response. However, breathing is irregular, and the extremities appear bluish while the trunk remains pink, suggesting peripheral cyanosis. These findings are used to calculate the APGAR score. What is the APGAR score in this newborn?

- a) 5
- b) 6
- c) 8
- d) 9

Q168. When is this milestone first achieved in the child?



- a) 3 months
- b) 10 months
- c) 15 months
- d) 2 years

Q169. A 2-month-old infant presents with persistent jaundice since birth, along with dark (turmeric-colored) urine and pale (clay-colored) stools. These findings are suggestive of conjugated hyperbilirubinemia, most likely due to biliary obstruction such as biliary atresia. On examination, hepatomegaly is present with an increased liver span. Early differentiation from other causes of neonatal cholestasis is essential for timely management. Among available investigations, one provides characteristic histological features and is considered most specific for diagnosis. Which of the following is the most specific investigation to establish the diagnosis?

- a) Liver function test
- b) Ultrasound abdomen
- c) Peroperative cholangiogram
- d) Liver biopsy

Q170. A 7-month-old infant is brought for routine developmental assessment. At this age, the child is expected to achieve several gross motor, fine motor, and social milestones. Typically, infants can sit with support, transfer objects from one hand to another, and show interest in mirror images. These milestones reflect normal neurological and motor development for this age group. However, certain advanced motor skills are not yet expected at this stage. Which of the following milestones is NOT likely to be present in this child?

- a) Sitting with support
- b) Walking with support
- c) Transferring objects
- d) Mirror play



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- Q171. A neonate presents with severe respiratory distress and has already received bag and mask ventilation. On examination, the abdomen is scaphoid, and chest X-ray shows mediastinal shift to the right side. These findings are highly suggestive of congenital diaphragmatic hernia with herniation of abdominal contents into the thoracic cavity. Bag and mask ventilation can worsen the condition by causing gastric distension and further lung compression. Immediate priority is to decompress the stomach and reduce intrathoracic pressure. What is the next best step in management?
- a) Immediate surgical intervention
 - b) Increase oxygenation
 - c) Put a nasogastric tube
 - d) CPAP
- Q172. A 4-year-old previously healthy child presents with sudden onset high-grade fever, severe throat pain, and rapidly progressive respiratory distress. The child is drooling saliva, unable to swallow, and is sitting in a forward-leaning (tripod) position with the neck hyperextended. This posture helps maintain airway patency. The acute onset, toxic appearance, and presence of drooling strongly suggest a life-threatening upper airway obstruction. Which of the following is the most likely diagnosis?
- a) Acute exacerbation of asthma
 - b) Pneumothorax
 - c) Laryngotracheobronchitis
 - d) Acute epiglottitis
- Q173. A 4-day-old preterm neonate is brought with features suggestive of a large patent ductus arteriosus (PDA), such as respiratory distress and signs of increased pulmonary blood flow. In preterm infants, the ductus arteriosus may remain open due to increased prostaglandin levels. Medical management aims to promote closure by inhibiting prostaglandin synthesis. Certain drugs achieve this by blocking cyclooxygenase (COX), thereby reducing prostaglandin levels and facilitating ductal closure. Which of the following drugs is most appropriate for closure of PDA in this patient?
- a) PGE1
 - b) PGE2
 - c) Indomethacin
 - d) Naproxen
- Q174. A 6-month-old infant presents with progressively increasing breathing difficulty over the past 2 days. The child is afebrile and on examination has bilateral wheezing on auscultation. Chest X-ray reveals hyperinflation of the lungs, suggesting air trapping. This clinical picture is typical of a viral lower respiratory tract infection affecting small airways, commonly seen in infants. Which of the following is the most likely diagnosis?
- a) Asthma
 - b) Croup
 - c) Bronchiolitis
 - d) Epiglottitis
- Q175. A child presents with alopecia, psoriatic-like dermatitis involving the genital and perioral regions, along with features of hypogonadism. These findings are characteristic of a nutritional deficiency affecting epithelial integrity and growth. The dermatitis is typically periorificial and acral, and may be associated with diarrhea and growth retardation. This constellation of features is classically seen in a trace element deficiency, particularly affecting rapidly dividing tissues. Which of the following deficiencies is most likely responsible for this condition?
- a) Copper deficiency



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- b) Iron deficiency
- c) Zinc deficiency
- d) Magnesium deficiency

Q176. An 11-year-old boy presents with localized thigh pain without any history of trauma. X-ray shows a small lytic lesion in the diaphysis surrounded by a zone of reactive sclerosis, which is characteristic of osteoid osteoma, a benign bone tumor. The pain is typically worse at night and shows dramatic relief with salicylates or NSAIDs due to prostaglandin production. CT scan is the best modality to identify the nidus clearly. It is also one of the most common benign bone tumors in children and adolescents. Which of the following statements is NOT true regarding this condition?

- a) Child's pain is likely to be relieved by Salicylates
- b) CT is the best investigation to see pathological lesion
- c) It is most common True Benign Bone Tumour
- d) Chemotherapy is treatment of Choice

Q177. A young patient presents to the emergency department following a shoulder injury, with the affected limb held in abduction and external rotation. On examination, there is weakness of elbow flexion and loss of sensation over the lateral aspect of the forearm. Elbow flexion is primarily mediated by the biceps brachii and brachialis muscles, while sensation over the lateral forearm is supplied by the lateral cutaneous nerve of the forearm. These functions are carried by a specific peripheral nerve arising from the lateral cord of the brachial plexus. Which of the following nerves is most likely injured?

- a) Radial Nerve
- b) Axillary Nerve
- c) Musculocutaneous Nerve
- d) Ulnar Nerve

Q178. Most common cause of Non Traumatic Avascular Necrosis:



- a) Steroid Induced
- b) Idiopathic
- c) Caisson's disease
- d) Chronic alcoholism

Q179. A 12 year old male is brought to the ED after injuring his forearm rollerblading. He fell onto his palm and noted pain and a deformity in his forearm. Examination revealed normal vital signs and findings limited to his left arm. His clavicle, shoulder, humerus, and hand were non-tender. He was reluctant to move his shoulder since his forearm was in a splint and sling. There was an obvious angulation at the mid-forearm. He



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could move all his fingers. No circulatory or sensory deficits were detected. Radiographs of his forearm were obtained. The most likely diagnosis is?



- a) Galeazzi's fracture
- b) Supracondylar fracture
- c) Monteggia's fracture
- d) Cole's fracture

Q180. A 7-year-old boy presents with acute onset pain in the right hip and restricted range of movements. On examination, the limb is held in abduction and external rotation, suggesting an effusion within the hip joint. Although the hemogram is normal, a mildly elevated ESR raises suspicion of an underlying inflammatory or early infective process. In children, distinguishing between transient synovitis and early septic arthritis is crucial. Imaging-guided aspiration helps confirm the diagnosis and relieves joint pressure. What is the next best line of management in this patient?

- a) I/V Antibiotics and Splintage
- b) Watchful observation
- c) USG guided aspiration
- d) Immediate Hospitalization

Q181. A 65-year-old man with a history of prolonged sun exposure as a farmer presents with a small, scaly erythematous lesion on the helix of the ear. The lesion is located on a sun-exposed area and has features suggestive of a premalignant condition. Histopathological examination reveals degeneration of dermal collagen (solar elastosis) along with atypia of squamous cells confined to the lower layers of the epidermis. Importantly, the atypia does not involve the full thickness of the epidermis and there is no invasion into the dermis. Which of the following is the most likely diagnosis?

- a) Actinic keratosis
- b) Bowen's disease
- c) Keratoacanthoma
- d) Seborrheic keratosis

Q182. A 25-year-old woman presents with a history of patchy hair loss, initially starting as a small bald patch and progressing to a larger area over time. She reports recent psychological stress, and there is a family history of similar episodes and autoimmune diseases. On examination, the scalp appears smooth and normal without scarring, inflammation, or scaling, with a few depigmented (white) hairs present. This pattern is characteristic of an autoimmune-mediated non-scarring alopecia. Which of the following is the most appropriate diagnosis and management?

- a) Trichotillomania – Antidepressants
- b) Alopecia Areata – Prednisolone
- c) Scarring Alopecia - Intralesional triamcinolone
- d) Tinea Capitis – Oral Griseofulvin

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Q183. An 8-month-old child presents with itchy, exudative skin lesions involving the face, palms, and soles. There is a history of similar complaints among siblings, suggesting a contagious condition. However, the presence of exudative (weeping) lesions indicates an eczematous dermatitis rather than a simple infestation. In infants, such lesions commonly represent infantile eczema (atopic dermatitis), which may involve the face and extensors. The primary pathology is inflammatory, requiring anti-inflammatory treatment rather than antiparasitic therapy. What is the most appropriate treatment in this patient?

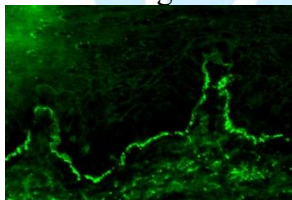
- a) Systemic ampicillin
- b) Topical betamethasone
- c) Systemic prednisolone
- d) Topical permethrin

Q184. A female patient presents with multiple warty lesions around vulva. The lesions are gradually increasing. On examination, they are soft sessile and did not bleed on touch. The likely diagnosis is:



- a) Condyloma acuminata
- b) Molluscum contagiosum
- c) Chancroid
- d) Herpes genitalis

Q185. Direct immune fluorescence skin biopsy performed on a 40 year old male patient who had multiple blisters over his trunk and extremities. The report states that immunofluorescence shows deposition of iga along the basement membrane. Which of the following diagnoses would be kept in mind with this result?



- a) Linear iga disorder
- b) Pemphigoid
- c) Dermatitis herpetiformis
- d) Pemphigus

Q186. A patient reports persistent sadness for the past 3 months, along with loss of interest in previously enjoyable activities. He also describes poor sleep, decreased appetite, and reduced energy levels affecting daily functioning. These symptoms are explored in detail, including their onset, duration, progression, and



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associated features. Such information helps in understanding the current illness and forming a clinical diagnosis. Which part of psychiatric history is primarily being elicited in this scenario?

- a) Past history
- b) Family history
- c) History of presenting illness
- d) Personal history

Q187. A 3-year-old child is brought with concerns of poor eye contact, delayed speech development, and limited social interaction with peers and family members. Parents also report that the child prefers repetitive behaviors, rigid routines, and shows resistance to change. There is lack of joint attention and difficulty in communication. These features are suggestive of a neurodevelopmental disorder affecting social communication and behavior. Which of the following is the most likely diagnosis?

- a) ADHD
- b) Autism spectrum disorder
- c) Intellectual disability
- d) Anxiety disorder

Q188. A patient presents with marked psychomotor abnormalities, including prolonged immobility and minimal response to external stimuli. On examination, the patient exhibits waxy flexibility, allowing limbs to remain in positions placed by the examiner. He also maintains unusual and rigid postures for extended periods. These features are characteristic of a specific subtype of schizophrenia involving prominent motor disturbances. Which of the following subtypes is most likely suggested in this case?

- a) Paranoid
- b) Catatonic
- c) Disorganized
- d) Residual

Q189. A 28-year-old man presents with a persistently elevated and expansive mood lasting for the past 7 days. He reports decreased need for sleep, increased energy, excessive goal-directed activity, and inflated self-esteem with grandiose ideas. These symptoms have led to significant impairment in social and occupational functioning. The duration and severity of symptoms indicate a full-blown mood episode rather than a milder form. Which of the following is the most likely diagnosis in this patient?

- a) Hypomania
- b) Mania
- c) Depression
- d) Cyclothymia

Q190. A patient presents with progressive cognitive decline characterized by a stepwise deterioration in memory and executive function. The history reveals multiple episodes of stroke, with each event followed by a noticeable decline in cognitive abilities. Neurological examination may show focal deficits corresponding to vascular insults. This pattern of decline is distinct from the gradual progression seen in degenerative dementias. Which of the following is the most likely diagnosis in this patient?

- a) Alzheimer's disease
- b) Vascular dementia
- c) Delirium
- d) Parkinson's dementia



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- Q191. A 60-year-old patient undergoes chest X-ray as part of evaluation for chronic cough. The radiograph demonstrates areas of increased radiopacity, appearing whiter compared to surrounding structures. On X-ray imaging, different tissues absorb varying amounts of radiation depending on their density. Structures with higher density absorb more X-rays and therefore appear more radiopaque (white) on the film. Among body tissues, one component shows the highest radiopacity due to its mineral content. Which of the following structures appears most radiopaque on X-ray?
- a) Air
 - b) Fat
 - c) Soft tissue
 - d) Bone
- Q192. A patient with clinical features of hyperthyroidism undergoes thyroid scintigraphy to assess functional activity of the gland. The scan demonstrates diffusely increased uptake throughout the entire thyroid gland, indicating uniform stimulation of thyroid tissue. This pattern is characteristic of an autoimmune condition in which antibodies stimulate the TSH receptors, leading to increased hormone production. It differs from focal uptake seen in nodular conditions. Which of the following is the most likely diagnosis?
- a) Thyroiditis
 - b) Graves' disease
 - c) Toxic adenoma
 - d) Thyroid carcinoma
- Q193. A CT scan of the abdomen at the level of the L1 vertebra reveals a major arterial branch arising from the anterior surface of the abdominal aorta. This vessel is responsible for supplying the foregut structures, including the stomach, liver, spleen, and proximal duodenum. It typically divides into three primary branches shortly after its origin. Its anatomical location and distribution are characteristic for identifying this vessel on imaging. Which of the following arteries is most likely being visualized?
- a) Superior mesenteric artery
 - b) Inferior mesenteric artery
 - c) Celiac trunk
 - d) Renal artery
- Q194. An elderly patient presents with gradually progressive headache, confusion, and cognitive decline occurring weeks after a minor head injury. There may be a history of trivial trauma that was initially ignored. CT scan of the brain reveals a crescent-shaped hypodense collection that crosses suture lines, indicating a slowly expanding bleed. This pattern is typical of venous bleeding due to rupture of bridging veins, commonly seen in elderly individuals with cerebral atrophy. Which of the following is the most likely diagnosis?
- a) Epidural hematoma
 - b) Subdural hematoma
 - c) Subarachnoid hemorrhage
 - d) Brain abscess
- Q195. A patient presents with abdominal distension, vomiting, and inability to pass stool or flatus. On examination, there may be abdominal tenderness and visible bowel loops. X-ray of the abdomen reveals multiple air-fluid levels along with dilated bowel loops. These findings indicate accumulation of gas and fluid proximal to a blockage in the intestinal tract. This radiological pattern is characteristic of a mechanical obstruction of the bowel. Which of the following is the most likely diagnosis?
- a) Perforation
 - b) Paralytic ileus



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- c) Intestinal obstruction
- d) Ascites

- Q196. A patient receiving intravenous fluid therapy is being closely monitored to assess adequacy of perfusion and renal function. Urine output is an important bedside parameter used to evaluate effective circulation and kidney function. Adequate urine output indicates sufficient renal perfusion and fluid balance. In adults, there is a standard minimum threshold below which oliguria is considered. What is the minimum acceptable urine output in adults?
- a) 0.2 mL/kg/hr
 - b) 0.5 mL/kg/hr
 - c) 1 mL/kg/hr
 - d) 2 mL/kg/hr
- Q197. A 30-year-old patient is planned for a short day-care surgical procedure and is induced with an inhalational anesthetic. The agent used produces rapid onset and quick recovery, allowing early discharge after surgery. This property is due to its low blood-gas solubility, resulting in faster equilibration between alveolar gas and blood. It is widely preferred for outpatient procedures because of its smooth induction and minimal irritation to airways. Which of the following anesthetic agents is most likely used in this case?
- a) Halothane
 - b) Isoflurane
 - c) Sevoflurane
 - d) Ether
- Q198. During a surgical procedure, there is an unexpected failure of oxygen supply from the central pipeline. Despite this, the patient continues to receive oxygen through an alternative source connected to the anesthesia machine. This backup system is essential to ensure uninterrupted oxygen delivery during emergencies. The anesthesia workstation is designed with a reserve source that can be immediately utilized when pipeline pressure drops. Which of the following components provides this backup oxygen supply?
- a) Flowmeter
 - b) Oxygen cylinder
 - c) Vaporizers
 - d) Rotameter
- Q199. During induction of anesthesia, the oxygen flush valve is accidentally activated, delivering a high flow of oxygen directly into the breathing circuit. This bypasses the vaporizers and flowmeters, resulting in rapid delivery of large volumes of gas at high pressure. If the patient's airway is closed or resistance is high, this sudden increase in pressure can be transmitted to the lungs. Such a situation can lead to overdistension of alveoli and potential lung injury. What is the most significant risk associated with this event?
- a) Hypoxia
 - b) Barotrauma
 - c) Hypercapnia
 - d) Hypotension
- Q200. During laryngoscopy, the anesthetist is able to visualize only the epiglottis, with no view of the glottic opening or vocal cords. This indicates a difficult airway where alignment of the oral, pharyngeal, and laryngeal axes is suboptimal. The Cormack–Lehane grading system is used to classify the laryngeal view during intubation and predict difficulty. Visualization limited to only the epiglottis corresponds to a specific grade in this classification. What is the Cormack–Lehane grade in this scenario?
- a) Grade I
 - b) Grade II
 - c) Grade III
 - d) Grade IV



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ANSWERS

1	b
2	c
3	b
4	b
5	b
6	b
7	b
8	c
9	b
10	b
11	b
12	b
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191	d
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193	c
194	b
195	c
196	b
197	c
198	b
199	b
200	c



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